

CS 57800 – Statistical Machine Learning

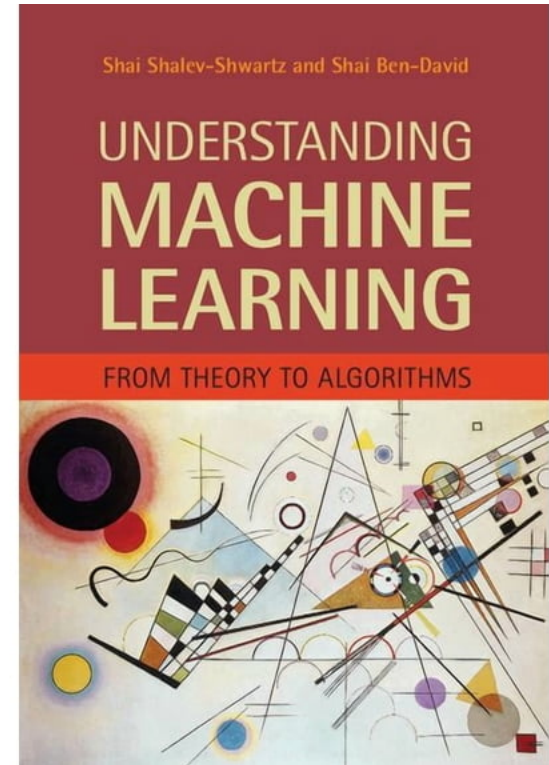
Lecture 2 Unsupervised Learning

https://ruizhezhang.com/course_fall_2026.html

Outline

Unsupervised Learning

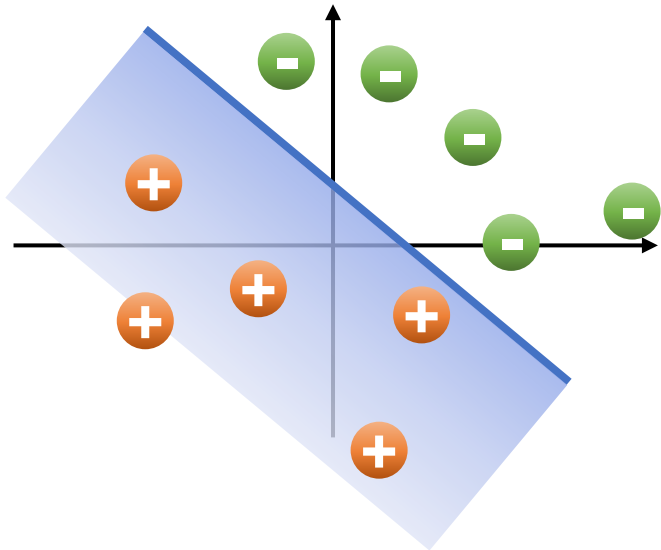
- Intro to Clustering
- K-Means
- Spectral Clustering
- ...



Unsupervised Learning

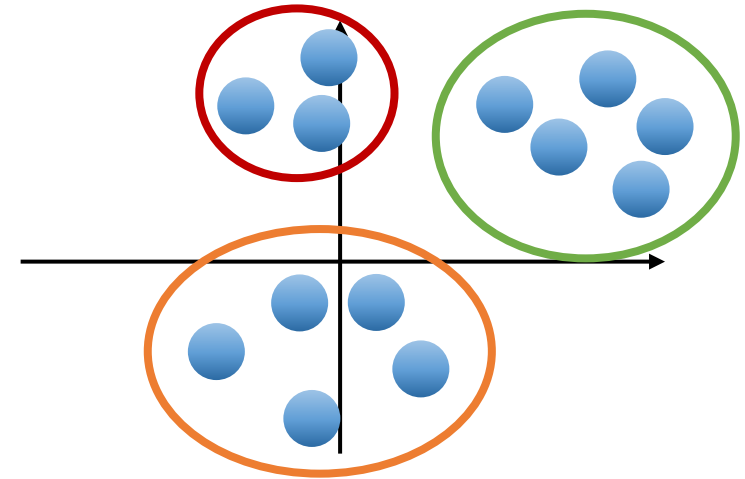
Supervised learning

- Training data has labels
- Make predictions for new data
- Measure success by population/empirical risk



Unsupervised learning

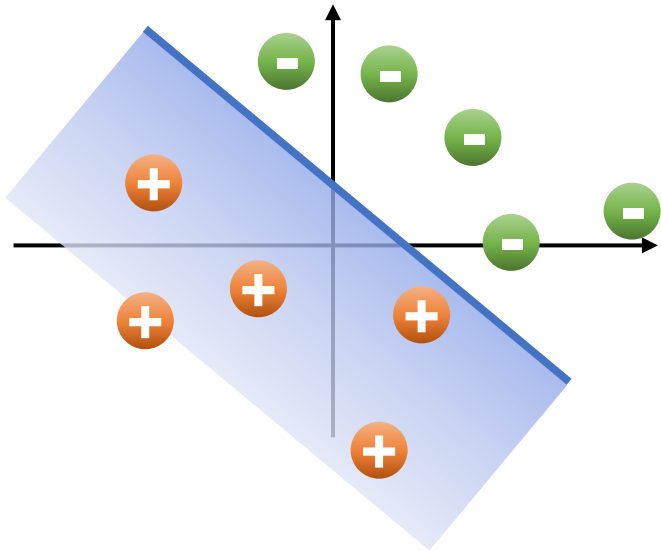
- Data without label
- Identify hidden structures in the data
- No “ground truth”



Unsupervised Learning

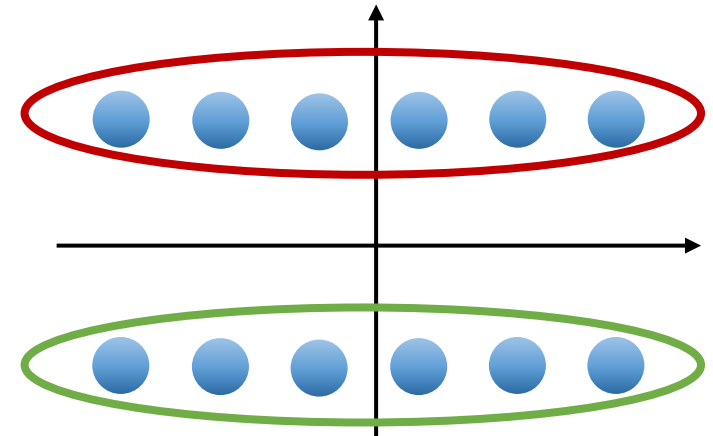
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Unsupervised learning

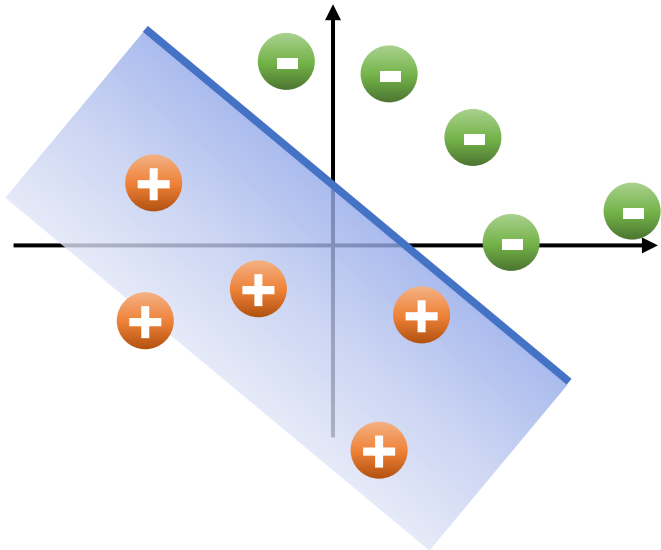
- Data without label
- Identify hidden structures in the data
- No “ground truth”
 - Not separate close-by points:



Unsupervised Learning

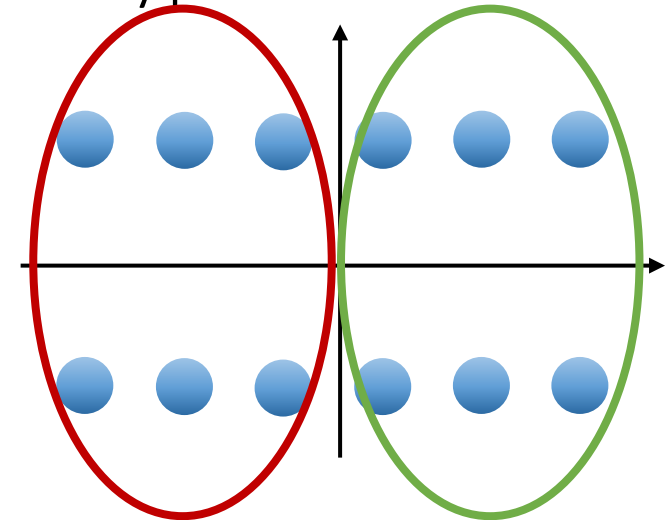
Supervised learning

- Training data has labels
- Make predictions for new data
- Measure success by population/empirical risk



Unsupervised learning

- Data without label
- Identify hidden structures in the data
- No “ground truth”
 - Not separate close-by points
 - No far-away points in the same cluster:



Clustering

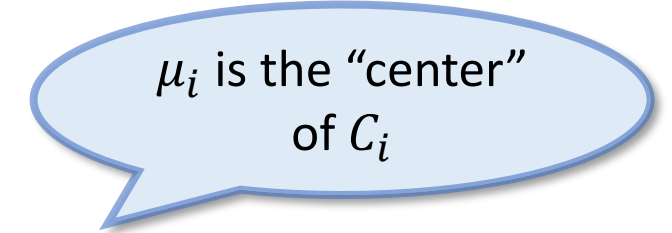
- **Input:** a set of elements $\mathcal{X}$, a distance (or similarity) function $d : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ that is symmetric, $d(x, x) = 0 \ \forall x \in \mathcal{X}$, and satisfies triangle inequality, (and a parameter k for the number of required clusters)
- **Output:** a partition of the domain set $C = (C_1, \dots, C_k)$ such that $\mathcal{X} = C_1 \cup \dots \cup C_k$ and $C_i \cap C_j = \emptyset \ \forall i \neq j$
- **Goal:** an **objective function** $G((\mathcal{X}, d), C)$ is minimized

Clustering

Center-based objectives:

- Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be a monotonic cost function
- Define the corresponding objective function:

$$G((\mathcal{X}, d), (C_1, \dots, C_k)) := \min_{\mu_1, \dots, \mu_k \in \mathcal{X}'} \sum_{i=1}^k \sum_{x \in C_i} f(d(x, \mu_i))$$



where $\mathcal{X}'$ can be $\mathcal{X}$ or some superset of $\mathcal{X}$ (such as $\mathbb{R}^d$)

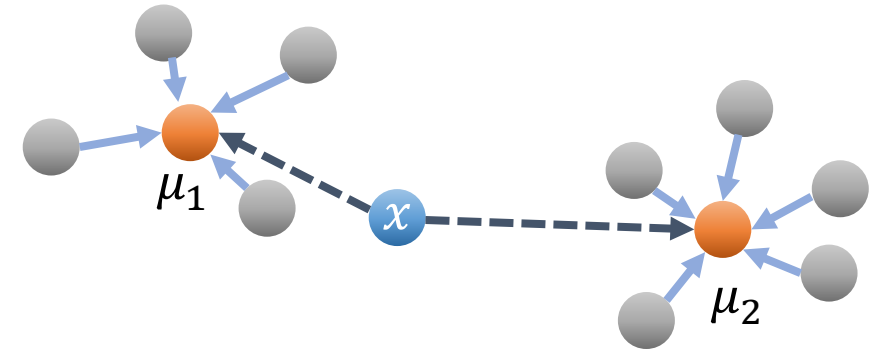
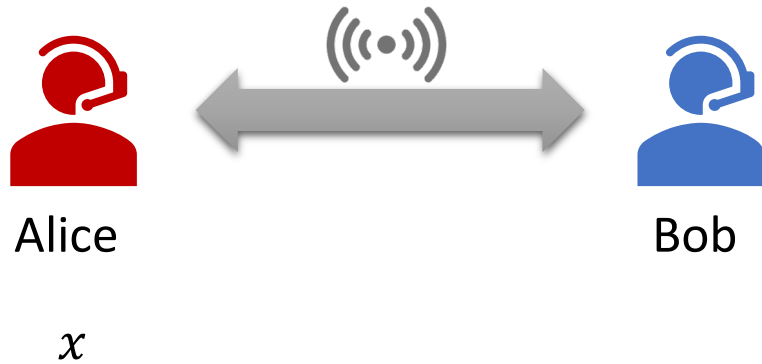
Example 1: k -means

$$G_{k\text{-means}}((\mathcal{X}, d), (C_1, \dots, C_k)) := \min_{\mu_1, \dots, \mu_k \in \mathcal{X}'} \sum_{i=1}^k \sum_{x \in C_i} d(x, \mu_i)^2$$

Clustering

Example 1: k -means

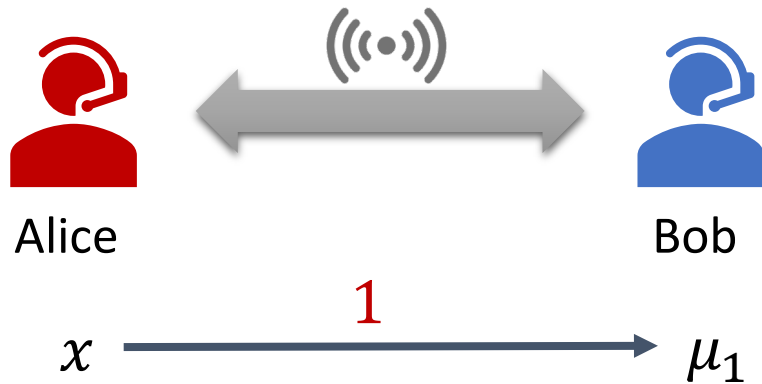
$$\mu_i := \operatorname{argmin}_{\mu \in \mathcal{X}'} \sum_{x \in C_i} d(x, \mu)^2, \quad G_{k\text{-means}} = \sum_{i=1}^k \sum_{x \in C_i} d(x, \mu_i)^2$$



Clustering

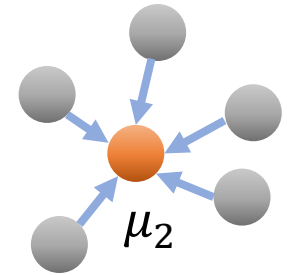
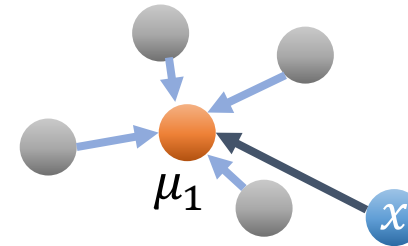
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$$\text{Distortion} = d(x, \mu_1)^2$$

Squared distance is a convention in signal processing, roughly the “energy” of the signal

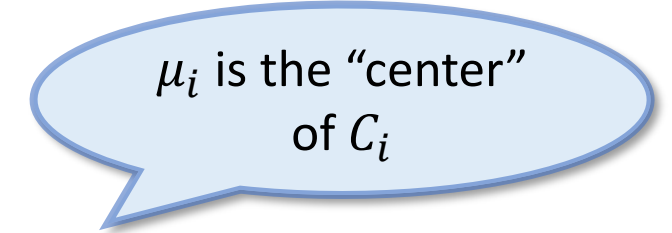


Clustering

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$$G((\mathcal{X}, d), (C_1, \dots, C_k)) := \min_{\mu_1, \dots, \mu_k \in \mathcal{X}'} \sum_{i=1}^k \sum_{x \in C_i} f(d(x, \mu_i))$$



where $\mathcal{X}'$ can be $\mathcal{X}$ or some superset of $\mathcal{X}$ (such as $\mathbb{R}^d$)

Example 2: k -medoids

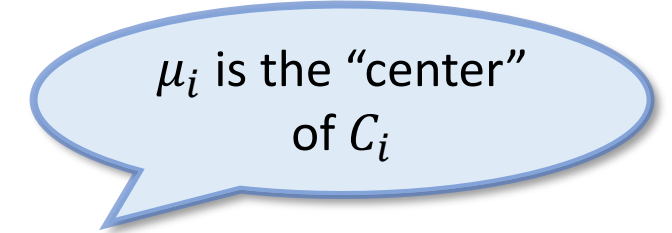
$$G_{k\text{-medoids}}((\mathcal{X}, d), (C_1, \dots, C_k)) := \min_{\mu_1, \dots, \mu_k \in \mathcal{X}} \sum_{i=1}^k \sum_{x \in C_i} d(x, \mu_i)^2$$

Clustering

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where $\mathcal{X}'$ can be $\mathcal{X}$ or some superset of $\mathcal{X}$ (such as $\mathbb{R}^d$)

Example 3: k -median

$$G_{k\text{-median}}((\mathcal{X}, d), (C_1, \dots, C_k)) := \min_{\mu_1, \dots, \mu_k \in \mathcal{X}} \sum_{i=1}^k \sum_{x \in C_i} d(x, \mu_i)$$

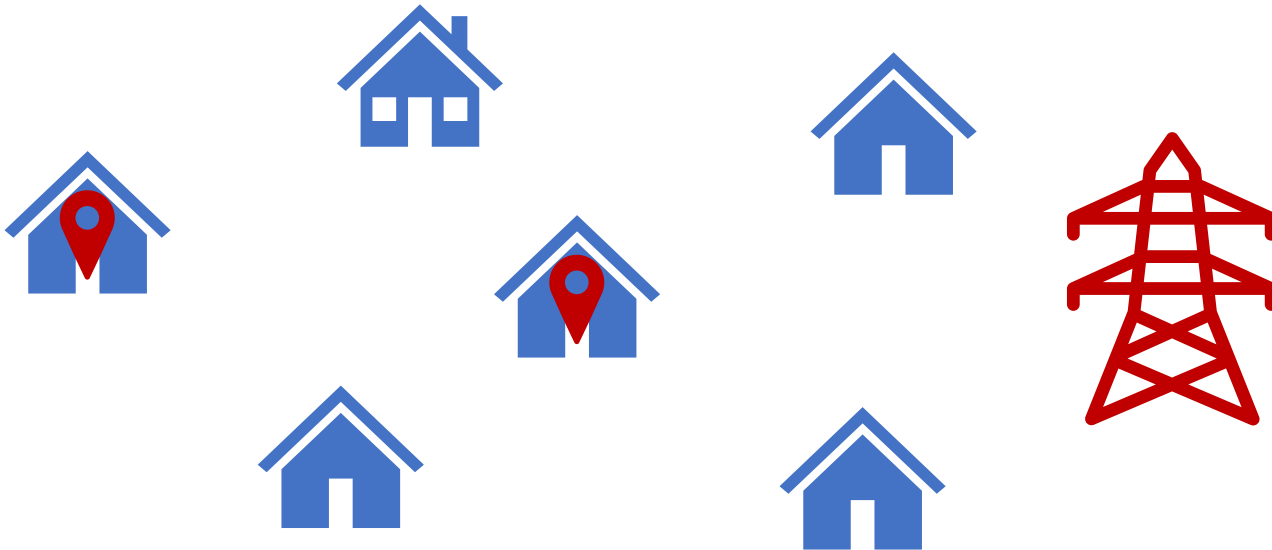
Clustering

Example 3: k -median

$$\mu_i := \operatorname{argmin}_{\mu \in C_i} \sum_{x \in C_i} d(x, \mu), \quad G_{k\text{-median}} = \sum_{i=1}^k \sum_{x \in C_i} d(x, \mu_i)$$

$$G_{k\text{-median}}^* = \min_{\mu_1, \dots, \mu_k \in \mathcal{X}} \sum_x \min_{i \in [k]} d(x, \mu_i)$$

Facility location problem:



$G_{k\text{-median}}$ is the average distance

Clustering

Example 3': k -median

$$\mu_i := \operatorname{argmin}_{\mu \in \mathcal{X}'} \sum_{x \in C_i} d(x, \mu), \quad G_{k\text{-median}} = \sum_{i=1}^k \sum_{x \in C_i} d(x, \mu_i)$$

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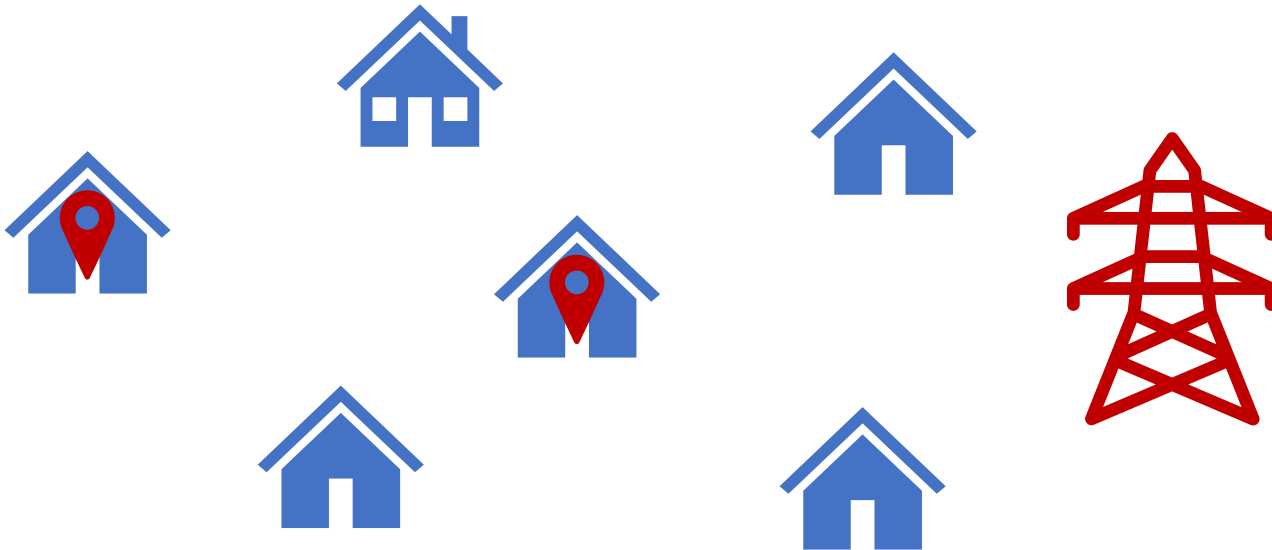
How about the worst distance?

Clustering

Example 4: k -center

$$G_{k\text{-center}}^* = \min_{\mu_1, \dots, \mu_k \in \mathcal{X}} \max_x \min_{i \in [k]} d(x, \mu_i)$$

Facility location problem:



$G_{k\text{-center}}$ is the worst distance

Clustering

- Different objective functions and different distance functions result in different algorithms and complexities
- Clustering has become a subfield in theoretical computer science and machine learning

Exact solutions:

- k -means, k -medoids, k -median, and k -center are **NP-hard**!

Approximate solutions:

- An α -approximation for a minimization problem returns a solution of cost at most $\alpha \cdot \text{OPT}$
- A $(c - \eta)$ -hardness of approximation means, for any constant $\eta > 0$, there is no polynomial time $(c - \eta)$ -approximation algorithm under certain complexity assumptions

Clustering

$\mu_i \in \mathcal{X}$, general metric space

Problems	Best poly-time approx.	Best hardness of approx.	References
k -centers	2	$2 - \eta$	[Hsu-Nemhauser '79, Hochbaum-Shmoys '85,]
k -median	$2 + \epsilon$	$1.416 - \eta$	[Cohen-Addad-Grandoni-Lee-Schwiegelshohn-Svensson '25, Anand-Lee '24]
k -medoids	$4.9 + \epsilon$		[Anand-Charikar-Cohen-Addad-Gao-Grandoni-Lee-Sharma-Wijland '26]

Euclidean space

Problems	Best poly-time approx.	Best hardness of approx.	References
k -means	$3.693 + \epsilon$	$1.36 - \eta$	[Anand-Charikar-Cohen-Addad-Gao-Grandoni-Lee-Sharma-Wijland '26, Cohen-Addad-C. S.-Lee '21]
k -median	$2 + \epsilon$	$1.08 - \eta$	

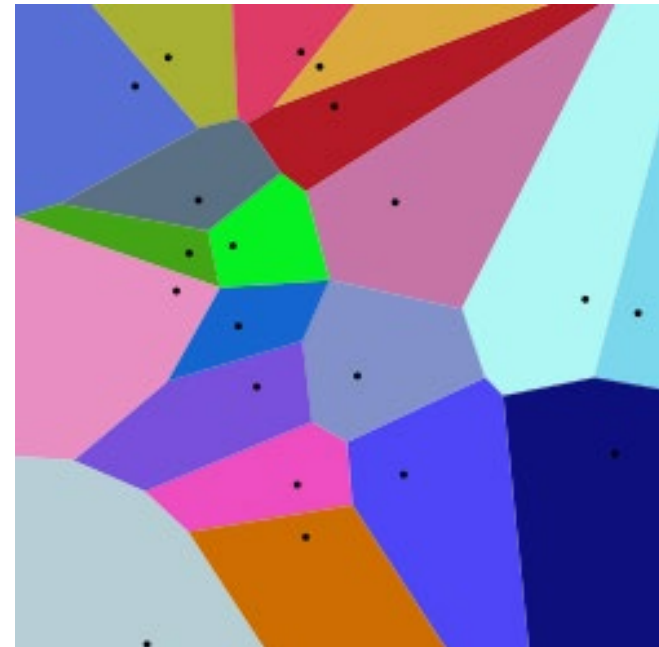
k -Means

Given m points $\mathcal{X} = \{x_1, \dots, x_m\} \subset \mathbb{R}^d$ and $k \in \mathbb{N}_+$, find $\{\mu_1, \dots, \mu_k\} \subset \mathbb{R}^d$ that minimizes the objective function

$$G(\mu_1, \dots, \mu_k) = \sum_{i=1}^m \min_{j \in [k]} \|x_i - \mu_j\|^2$$

- The partition $(C_1, \dots, C_k)$ of $\mathcal{X}$ is induced by the centers $(\mu_1, \dots, \mu_k)$
- Moreover, $(\mu_1, \dots, \mu_k)$ gives a **Voronoi partition** of $\mathbb{R}^d$:
 - $\mathbb{R}^d$ is decomposed into k convex cells $C_1, \dots, C_k$
 - We can recover the previous form of the objective function:

$$G(C_{1:k}; \mu_{1:k}) = \sum_{j=1}^k \sum_{x_i \in C_j} d(x_i, \mu_j)^2$$



k -Means

Lemma (One cluster). For any set $C \subset \mathbb{R}^d$ and any $\mu \in \mathbb{R}^d$,

$$G(C; \mu) = G(C; \bar{C}) + |C| \cdot \|\mu - \bar{C}\|^2, \quad \bar{C} := \frac{1}{|C|} \sum_{x \in C} x$$

Proof.

- Consider the following **bias-variance decomposition** of random vectors:

$$\mathbb{E}[\|X - \mu\|^2] = \mathbb{E}[\|X - \mathbb{E}[X]\|^2] + \|\mu - \mathbb{E}[X]\|^2, \quad \forall \mu \in \mathbb{R}^d$$

- Define X to be a uniformly random point in C
- Then, $\mathbb{E}[X] = \bar{C}$, and

$$\mathbb{E}[\|X - \mu\|^2] = \frac{1}{|C|} \sum_{x \in C} \|x - \mu\|^2 = \frac{1}{|C|} G(C; \mu)$$

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- For a single cluster, the optimal choice for μ is the **centroid** of C

k -Means

k -means algorithm:

1. Initialize centers $\mu_1, \dots, \mu_k \in \mathbb{R}^d$ and clusters $C_1, \dots, C_k$
2. Repeat until there is no further change in the objective function:
 - i. For each $j \in [k]$: $C_j \leftarrow \{x \in \mathcal{X} \text{ whose closest center is } \mu_j\}$
 - ii. For each $j \in [k]$: $z_j \leftarrow \overline{C_j}$

Lemma (Convergence). During the k -means algorithm, the cost monotonically decreases.

Proof.

- Let $\mu_1^{(t)}, \dots, \mu_k^{(t)}, C_1^{(t)}, \dots, C_k^{(t)}$ denote the centers and clusters at the start of the t -th iteration

$$G\left(C_{1:k}^{(t+1)}; \mu_{1:k}^{(t+1)}\right) \leq G\left(C_{1:k}^{(t+1)}; \mu_{1:k}^{(t)}\right) \leq G\left(C_{1:k}^{(t)}; \mu_{1:k}^{(t)}\right)$$

(2.ii.) (2.i.)



k -Means

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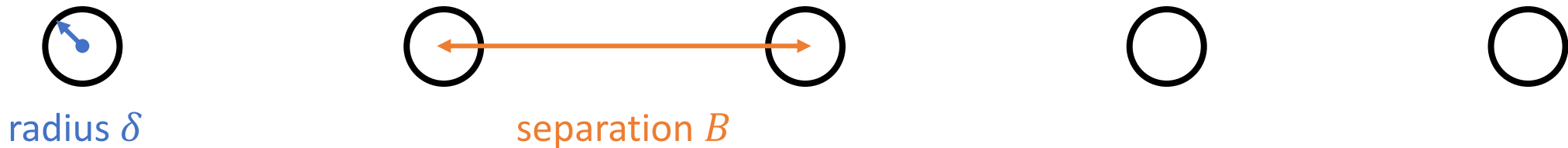
➤ The k -means algorithm will converge to a **stationary point** of the objective function

k -Means

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5-cluster example:



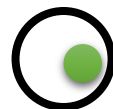
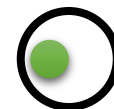
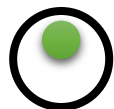
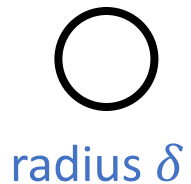
$$G_{\text{OPT}} \approx \delta^2 n$$

k -Means

k -means algorithm:

1. Initialize centers $\mu_1, \dots, \mu_k \in \mathbb{R}^d$ and clusters $C_1, \dots, C_k$
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5-cluster example:



separation B

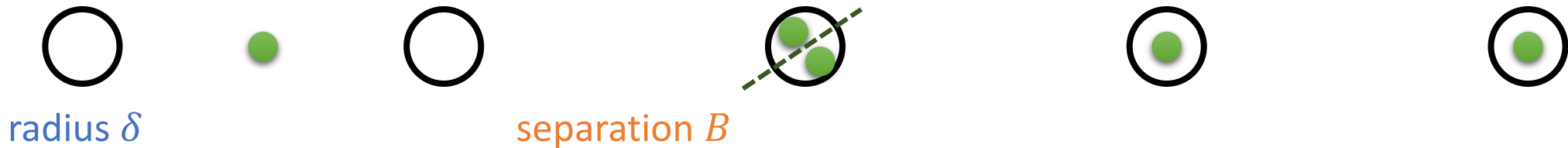
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k -Means

k -means algorithm:

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5-cluster example:



$$G_{\text{OPT}} \approx \delta^2 n \quad \ll \quad G_{\text{Local}} \approx \Omega(B^2 n)$$

k -Means++

k -means++ initialization:

1. Sample $\mu_1 \in \mathcal{X}$ uniformly at random
2. For each $j \in \{2, \dots, k\}$:
 - i. Sample $\mu_j = x \in \mathcal{X}$ with probability $\propto G(x; \mu_{1:j-1}) := \min_{i < j} \|x - \mu_i\|^2$

Theorem (Arthur-Vassilvitskii '07). Let $\mu_1, \dots, \mu_k \in \mathcal{X}$ be the output of k -means++ and $\mu_1^*, \dots, \mu_k^*$ be the optimal centers. Then,

$$\mathbb{E}[G(\mathcal{X}; \mu_{1:k})] \leq O(\log k) \cdot G(\mathcal{X}; \mu_{1:k}^*)$$

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Lemma (One cluster). For any set $C \subset \mathbb{R}^d$ and any $\mu \in \mathbb{R}^d$,

$$G(C; \mu) = G(C; \text{mean}(C)) + |C| \cdot \|\mu - \text{mean}(C)\|^2, \quad \text{mean}(C) := \frac{1}{|C|} \sum_{x \in C} x$$

By averaging over $\mu \in C$, we have

$$\mathbb{E}_{\mu \sim C}[G(C; \mu)] = G(C; \text{mean}(C)) + \sum_{\mu \in C} \|\mu - \text{mean}(C)\|^2 = 2G(C; \text{mean}(C)) = \mathbf{2OPT}(C)$$

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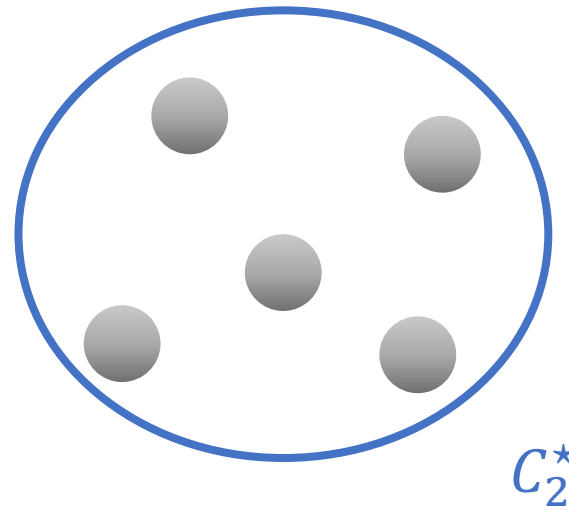
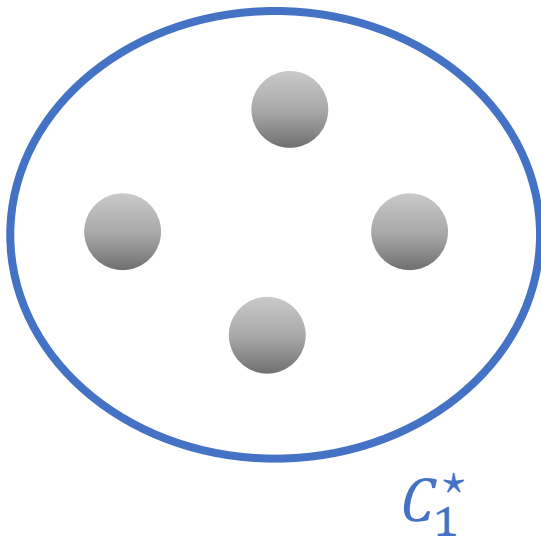
$$\mathbb{E}_{\mu \sim C}[G(C; \mu)] = 2\text{OPT}(C)$$

- Let $C_1^*, \dots, C_k^*$ be the optimal clusters induced by $\mu_1^*, \dots, \mu_k^*$
- Conditioned on $\mu_1 \in C_i^*$, we have μ_1 is uniformly sampled from C_i^* , and
$$\mathbb{E}_{\mu_1 \sim C_i}[G(C_i^*; \mu_1)] = 2G(C_i^*; \mu_1^*)$$
- So, the first center is good

k -Means++

k -means++ initialization:

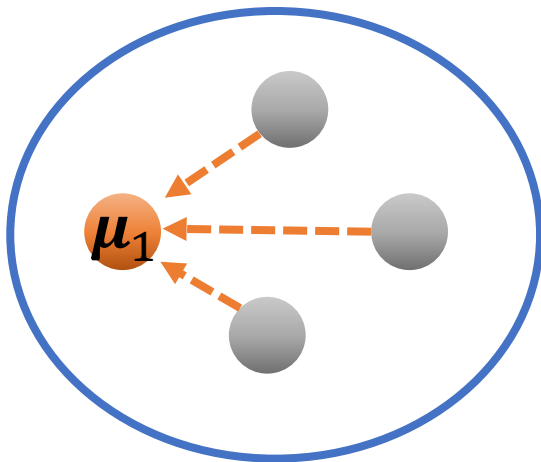
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k -Means++

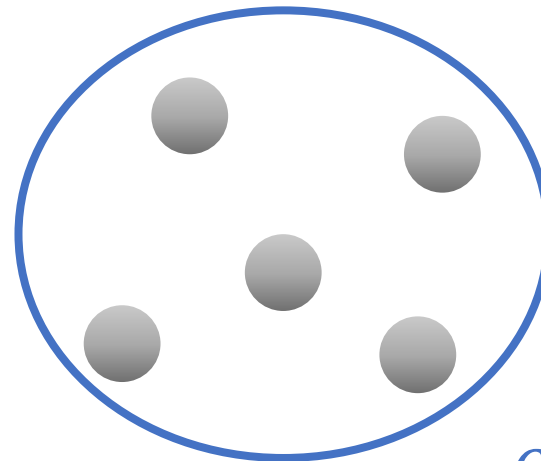
k -means++ initialization:

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$$G(C_1^*; \mu_1) \approx 2G(C_1^*; \mu_1^*)$$

C_1^*

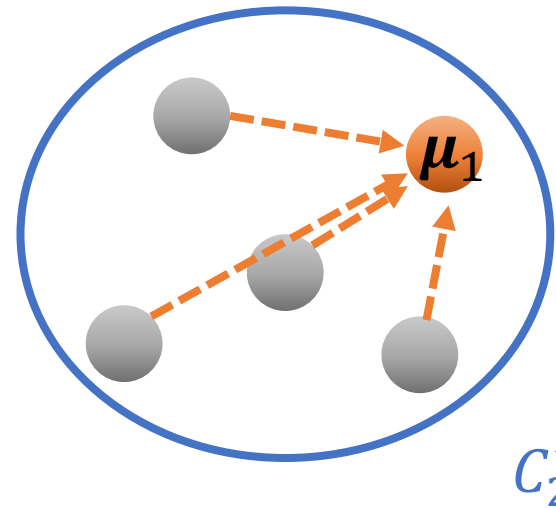
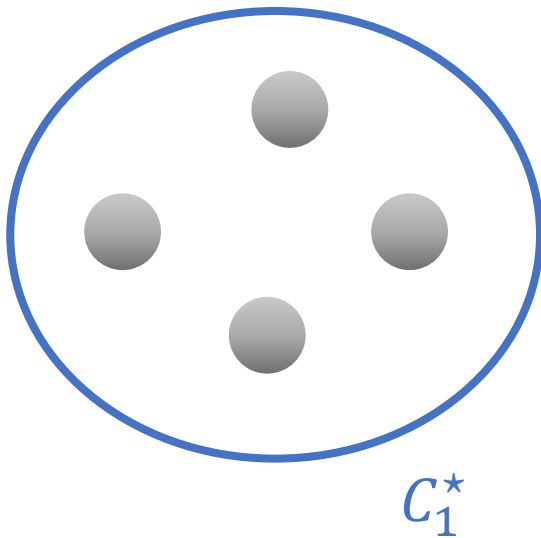


C_2^*

k -Means++

k -means++ initialization:

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$$G(C_2^*; \mu_2) \approx 2G(C_2^*; \mu_2^*)$$

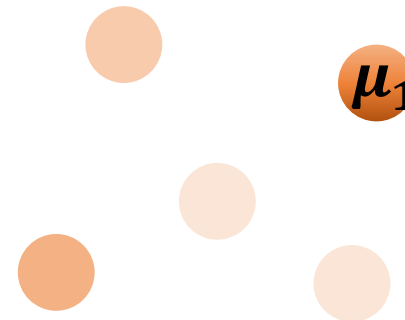
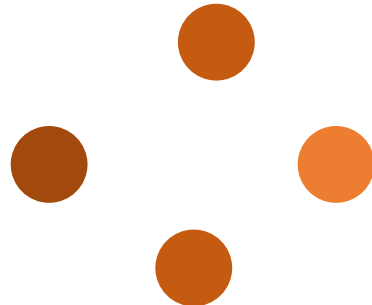
k -Means++

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 - i. Sample $\mu_j = x \in \mathcal{X}$ with probability $\propto G(x; \mu_{1:j-1}) := \min_{i < j} \|x - \mu_i\|^2$

- Later centers are **not** uniform within their clusters
- The sampling distribution is biased towards points far away from the current centers
- Nevertheless, we can prove that this biased choice will not select a terrible center

Points in the same cluster have similar sampling probability



sampling prob.

0

1



k -Means++

k -means++ initialization:

1. Sample $\mu_1 \in \mathcal{X}$ uniformly at random
2. For each $j \in \{2, \dots, k\}$:
 - i. Sample $\mu_j = x \in \mathcal{X}$ with probability $\propto G(x; \mu_{1:j-1}) := \min_{i < j} \|x - \mu_i\|^2$

Lemma. In the j -th iteration, let μ_j be the sampled center. Then, for any $i \in [k]$,

$$\mathbb{E}[G(C_i^*; \mu_{1:j}) | \mu_j \in C_i^*] \leq 8G(C_i^*; \mu_i^*)$$

Proof.

- Conditioned on $\mu_j \in C_i^*$, for any $x \in C_i^*$,

$$\Pr[\mu_j = x | \mu_j \in C_i^*] = \frac{G(x; \mu_{1:j-1})}{G(C_i^*; \mu_{1:j-1})}$$

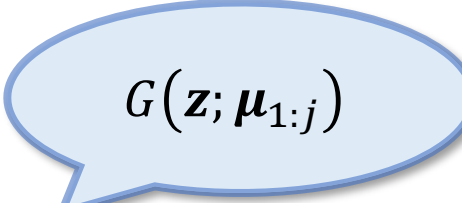
k -Means++

Lemma. In the j -th iteration, let μ_j be the sampled center. Then, for any $i \in [k]$,

$$\mathbb{E}[G(C_i^*; \mu_{1:j}) | \mu_j \in C_i^*] \leq 8G(C_i^*; \mu_i^*)$$

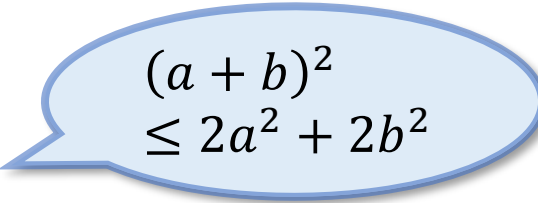
Proof.

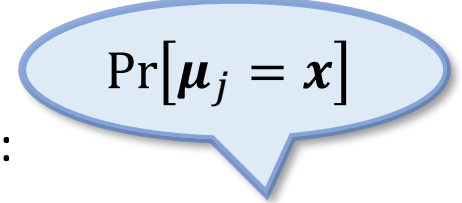
$$\mathbb{E}[G(C_i^*; \mu_{1:j}) | \mu_j \in C_i^*] = \sum_{x \in C_i^*} \frac{G(x; \mu_{1:j-1})}{G(C_i^*; \mu_{1:j-1})} \cdot \sum_{z \in C_i^*} \min\{G(z; \mu_{1:j-1}), \|z - x\|^2\}$$


$$G(z; \mu_{1:j})$$

- For $G(x; \mu_{1:j-1})$, pick any $y \in C_i^*$ and let $\mu_y \in \{\mu_1, \dots, \mu_{j-1}\}$ be the closest center to y

$$\begin{aligned} G(x; \mu_{1:j-1}) &\leq \|x - \mu_y\|^2 \leq (\|x - y\| + \|y - \mu_y\|)^2 \\ &\leq 2\|x - y\|^2 + 2\|y - \mu_y\|^2 \\ &= 2\|x - y\|^2 + 2G(y; \mu_{1:j-1}) \end{aligned}$$


$$\begin{aligned} (a + b)^2 \\ \leq 2a^2 + 2b^2 \end{aligned}$$


$$\Pr[\mu_j = x]$$

- Averaging over $y \in C_i^*$:

$$G(x; \mu_{1:j-1}) \leq \frac{2}{|C_i^*|} (G(C_i^*; x) + G(C_i^*; \mu_{1:j-1}))$$

k -Means++

Lemma. In the j -th iteration, let μ_j be the sampled center. Then, for any $i \in [k]$,

$$\mathbb{E}[G(C_i^*; \mu_{1:j}) | \mu_j \in C_i^*] \leq 8G(C_i^*; \mu_i^*)$$

Proof.

$$\begin{aligned} \mathbb{E}[G(C_i^*; \mu_{1:j}) | \mu_j \in C_i^*] &= \sum_{x \in C_i^*} \frac{G(x; \mu_{1:j-1})}{G(C_i^*; \mu_{1:j-1})} \cdot \sum_{z \in C_i^*} \min\{G(z; \mu_{1:j-1}), \|z - x\|^2\} \\ &\leq \frac{2}{|C_i^*|} \left(\sum_{x \in C_i^*} \frac{G(C_i^*; x)}{G(C_i^*; \mu_{1:j-1})} \cdot \sum_{z \in C_i^*} G(z; \mu_{1:j-1}) + \sum_{x \in C_i^*} \frac{G(C_i^*; \mu_{1:j-1})}{G(C_i^*; \mu_{1:j-1})} \cdot \sum_{z \in C_i^*} \|z - x\|^2 \right) \end{aligned}$$

$$G(x; \mu_{1:j-1}) \leq \frac{2}{|C_i^*|} (G(C_i^*; x) + G(C_i^*; \mu_{1:j-1}))$$

k -Means++

Lemma. In the j -th iteration, let $\boldsymbol{\mu}_j$ be the sampled center. Then, for any $i \in [k]$,

$$\mathbb{E}[G(C_i^*; \boldsymbol{\mu}_{1:j}) | \boldsymbol{\mu}_j \in C_i^*] \leq 8G(C_i^*; \boldsymbol{\mu}_i^*)$$

Proof.

$$\begin{aligned} \mathbb{E}[G(C_i^*; \boldsymbol{\mu}_{1:j}) | \boldsymbol{\mu}_j \in C_i^*] &= \sum_{\mathbf{x} \in C_i^*} \frac{G(\mathbf{x}; \boldsymbol{\mu}_{1:j-1})}{G(C_i^*; \boldsymbol{\mu}_{1:j-1})} \cdot \sum_{\mathbf{z} \in C_i^*} \min\{G(\mathbf{z}; \boldsymbol{\mu}_{1:j-1}), \|\mathbf{z} - \mathbf{x}\|^2\} \\ &\leq \frac{2}{|C_i^*|} \left(\sum_{\mathbf{x} \in C_i^*} \frac{G(C_i^*; \mathbf{x})}{G(C_i^*; \boldsymbol{\mu}_{1:j-1})} \cdot \sum_{\mathbf{z} \in C_i^*} G(\mathbf{z}; \boldsymbol{\mu}_{1:j-1}) + \sum_{\mathbf{x} \in C_i^*} \frac{G(C_i^*; \boldsymbol{\mu}_{1:j-1})}{G(C_i^*; \boldsymbol{\mu}_{1:j-1})} \cdot \sum_{\mathbf{z} \in C_i^*} \|\mathbf{z} - \mathbf{x}\|^2 \right) \\ &= \frac{4}{|C_i^*|} \sum_{\mathbf{x} \in C_i^*} G(C_i^*; \mathbf{x}) = 4\mathbb{E}_{\mathbf{x} \sim C_i^*}[G(C_i^*; \mathbf{x})] = 8G(C_i^*; \boldsymbol{\mu}_i^*) \end{aligned}$$

■

k -Means++

k -means++ initialization:

1. Sample $\mu_1 \in \mathcal{X}$ uniformly at random
2. For each $j \in \{2, \dots, k\}$:
 - i. Sample $\mu_j = x \in \mathcal{X}$ with probability $\propto G(x; \mu_{1:j-1}) := \min_{i < j} \|x - \mu_i\|^2$

Lemma. In the j -th iteration, let μ_j be the sampled center. Then, for any $i \in [k]$,

$$\mathbb{E}[G(C_i^*; \mu_{1:j}) | \mu_j \in C_i^*] \leq 8G(C_i^*; \mu_i^*)$$

➤ Why can't it achieve 8-approximation?

k -Means++

k -means++ initialization:

1. Sample $\mu_1 \in \mathcal{X}$ uniformly at random
2. For each $j \in \{2, \dots, k\}$:
 - i. Sample $\mu_j = x \in \mathcal{X}$ with probability $\propto G(x; \mu_{1:j-1}) := \min_{i < j} \|x - \mu_i\|^2$

- For any $t \leq k$, define the set of clusters that have been “hit”:

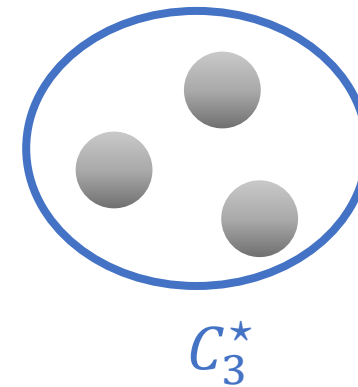
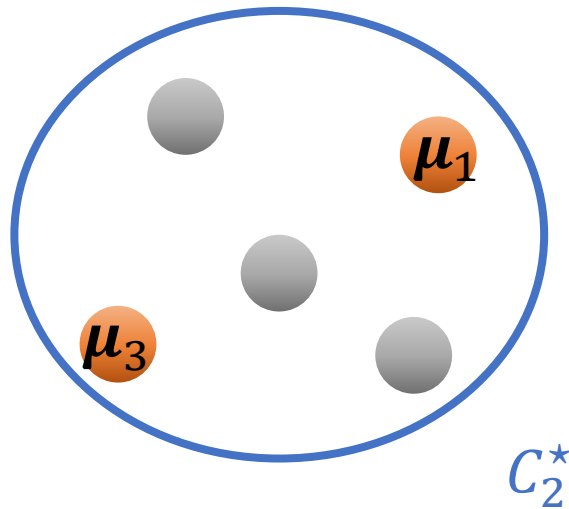
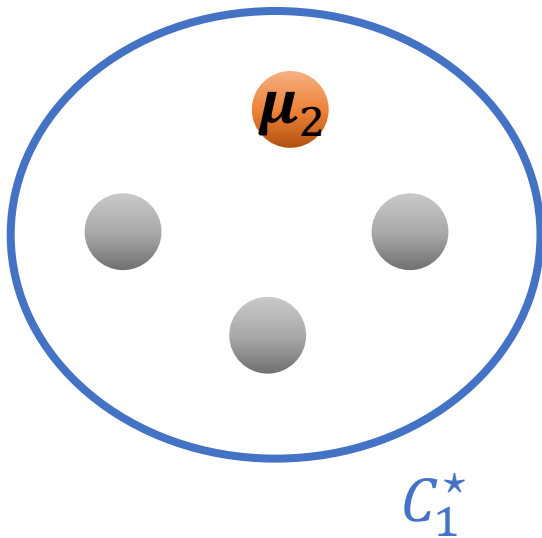
$$H_t := \{i \in [k] : C_i^* \cap \mu_{1:t} \neq \emptyset\}, \quad C_{H_t}^* := \bigcup_{i \in H_t} C_i^*$$

- Let $U_t := [k] \setminus H_t$ be the remaining “uncovered” clusters, and $W_t := t - |H_t|$ be the number of “wasted” iterations

k -Means++

k -means++ initialization:

1. Sample $\mu_1 \in \mathcal{X}$ uniformly at random
2. For each $j \in \{2, \dots, k\}$:
 - i. Sample $\mu_j = x \in \mathcal{X}$ with probability $\propto G(x; \mu_{1:j-1}) := \min_{i < j} \|x - \mu_i\|^2$



- $t = 3$
- $H_3 = \{1, 2\}$
- $U_3 = \{3\}$
- $W_3 = 1$

k -Means++

We claim that $\mathbb{E}[G(C_{H_t}^*; \boldsymbol{\mu}_{1:t})] \leq 8G(\mathcal{X}; \boldsymbol{\mu}_{1:k}^*)$ for any $t \leq k$

- Note that

$$G(C_{H_t}^*; \boldsymbol{\mu}_{1:t}) = \sum_{i \in H_t} G(C_i^*; \boldsymbol{\mu}_{1:t})$$

- Let $s_i \in [t]$ be the **first time** that C_i^* was “hit”. Then $G(C_i^*; \boldsymbol{\mu}_{1:t}) \leq G(C_i^*; \boldsymbol{\mu}_{1:s_i}) = G(C_i^*; \boldsymbol{\mu}_{s_i})$

$$\mathbb{E}[G(C_i^*; \boldsymbol{\mu}_{1:t})] \leq \sum_{s=1}^k \mathbb{E}[\mathbf{1}_{s_i=s} G(C_i^*; \boldsymbol{\mu}_s)] = \sum_{s=1}^k \mathbb{E} \left[\mathbb{E}_{\mu_s} [\mathbf{1}_{s_i=s} G(C_i^*; \boldsymbol{\mu}_s) | \boldsymbol{\mu}_{1:s-1}] \right]$$

- $\mathbf{1}_{s_i=s}$ is equivalent to $\mathbf{1}\{i \notin H_{s-1}\} \cdot \mathbf{1}\{\boldsymbol{\mu}_s \in C_i^*\}$; thus, we have

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{s_i=s} G(C_i^*; \boldsymbol{\mu}_s) | \boldsymbol{\mu}_{1:s-1}] &= \mathbf{1}\{i \notin H_{s-1}\} \cdot \mathbb{E}[G(C_i^*; \boldsymbol{\mu}_s) | \boldsymbol{\mu}_{1:s-1}, \boldsymbol{\mu}_s \in C_i^*] \cdot \Pr[\boldsymbol{\mu}_s \in C_i^* | \boldsymbol{\mu}_{1:s-1}] \\ &\leq 8G(C_i^*; \boldsymbol{\mu}_i^*) \cdot \mathbf{1}\{i \notin H_{s-1}\} \cdot \Pr[\boldsymbol{\mu}_s \in C_i^* | \boldsymbol{\mu}_{1:s-1}] \\ &= 8G(C_i^*; \boldsymbol{\mu}_i^*) \cdot \Pr[s_i = s | \boldsymbol{\mu}_{1:s-1}] \end{aligned}$$

k -Means++

We claim that $\mathbb{E}[G(C_{H_t}^*; \boldsymbol{\mu}_{1:t})] \leq 8G(\mathcal{X}; \boldsymbol{\mu}_{1:k}^*)$ for any $t \leq k$

- Therefore,

$$\begin{aligned}\mathbb{E}[G(C_i^*; \boldsymbol{\mu}_{1:t})] &\leq \sum_{s=1}^k \mathbb{E}_{\mu_{1:s-1}}[8G(C_i^*; \boldsymbol{\mu}_i^*) \cdot \Pr[s_i = s | \boldsymbol{\mu}_{1:s-1}]] \\ &= 8G(C_i^*; \boldsymbol{\mu}_i^*) \cdot \sum_{s=1}^k \Pr[s_i = s] \leq 8G(C_i^*; \boldsymbol{\mu}_i^*)\end{aligned}$$

- Thus, $\mathbb{E}[G(C_{H_t}^*; \boldsymbol{\mu}_{1:t})] \leq \sum_{i \in H_t} 8G(C_i^*; \boldsymbol{\mu}_i^*) \leq 8G(\mathcal{X}; \boldsymbol{\mu}_{1:k}^*)$

k -Means++

To analyze those uncovered clusters, we define a **potential function**

$$\Phi_t := W_t \frac{G(C_{U_t}^*; \mu_{1:t})}{|U_t|}$$

Average cost of an uncovered cluster

- Initial potential $t = 0$:

$$W_0 = 0, \quad \Phi_0 = 0$$

- Final potential $t = k$:

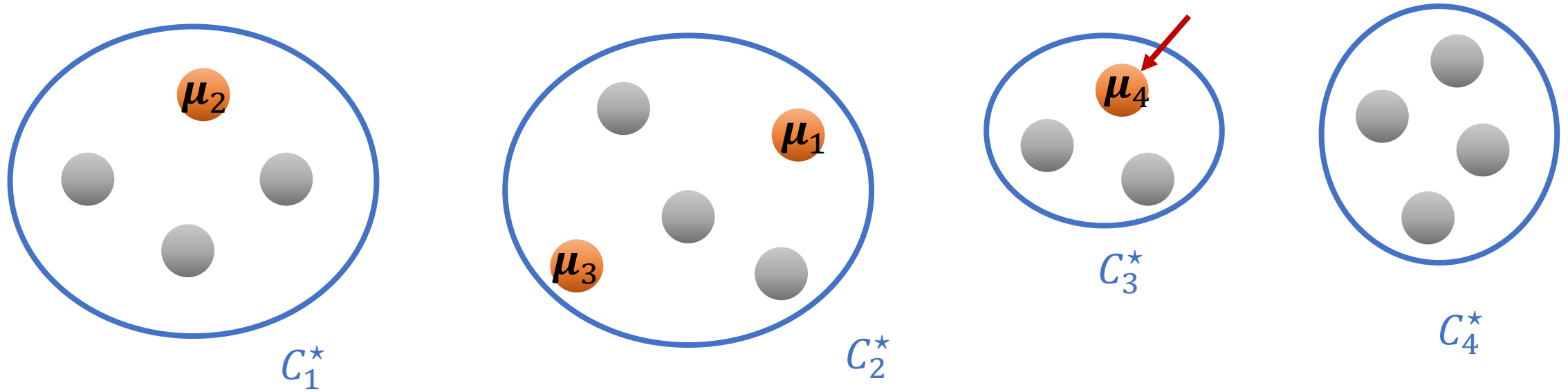
$$W_k = |U_k|, \quad \Phi_k = G(C_{U_k}^*; \mu_{1:k})$$

- Thus,

$$G(\mathcal{X}; \mu_{1:k}) = G(C_{H_k}^*; \mu_{1:k}) + \sum_{t=0}^{k-1} (\Phi_{t+1} - \Phi_t)$$

$\leq 8G(\mathcal{X}; \mu_{1:k}^*)$
in expectation

k -Means++



Good iteration

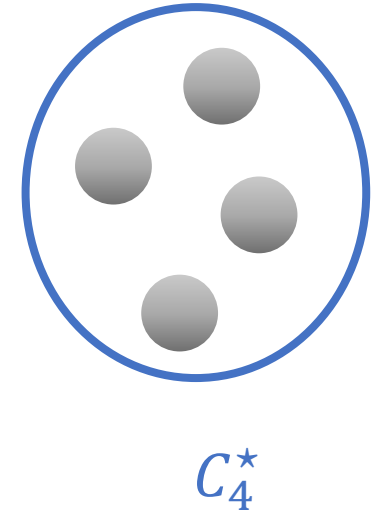
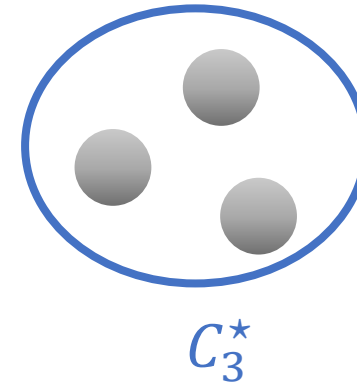
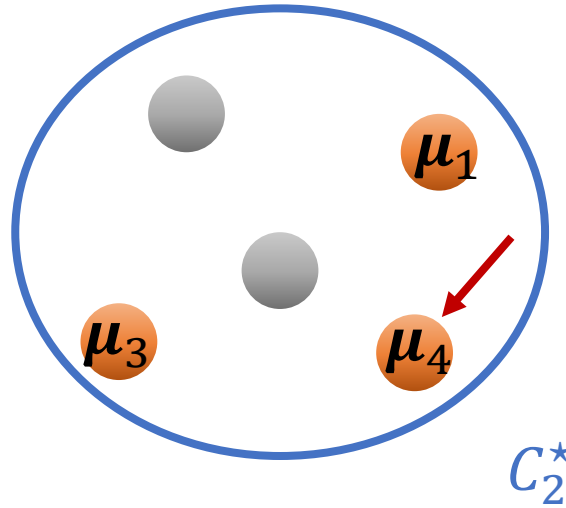
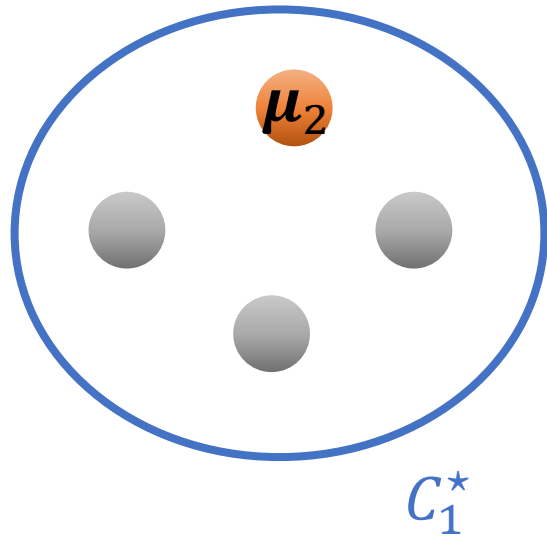
- $t = 3$
- $H_3 = \{1,2\}$
- $U_3 = \{3,4\}$
- $W_3 = 1$



- $t = 4$
- $H_4 = \{1,2,3\}$
- $U_4 = \{4\}$
- $W_4 = 1$

$$\Phi_4 = W_4 \frac{G(C_{U_4}^*; \mu_{1:4})}{|U_4|} = W_3 \frac{G(C_{U_3}^*; \mu_{1:4}) - G(C_3^*; \mu_4)}{|U_3| - 1}$$

k -Means++



Bad iteration

- $t = 3$
- $H_3 = \{1,2\}$
- $U_3 = \{3,4\}$
- $W_3 = 1$



- $t = 4$
- $H_4 = \{1,2\}$
- $U_4 = \{3,4\}$
- $W_4 = 2$

$$\Phi_4 = W_4 \frac{G(C_{U_4}^*; \mu_{1:4})}{|U_4|} = (W_3 + 1) \frac{G(C_{U_3}^*; \mu_{1:4})}{|U_3|}$$

k -Means++

Lemma (Good iteration). Let $t \leq k - 1$. Conditioned on $\mu_{1:t}$ and on the event that μ_{t+1} lands in an uncovered cluster, $\mathbb{E}[\Phi_{t+1}] \leq \Phi_t$

Proof.

- Suppose $\mu_{t+1} \in C_I^*$ with $I \in U_t$. Then, $W_{t+1} = W_t$ and $U_{t+1} = U_t \setminus \{I\}$

$$\Phi_{t+1} = W_{t+1} \frac{G(C_{U_{t+1}}^*; \mu_{1:t+1})}{|U_{t+1}|} \leq W_t \frac{G(C_{U_{t+1}}^*; \mu_{1:t})}{|U_{t+1}|} = W_t \frac{G(C_{U_t}^*; \mu_{1:t}) - G(C_I^*; \mu_{1:t})}{|U_t| - 1}$$

- Conditioned on landing in an uncovered cluster, we have for any $i \in U_t$,

$$\Pr[I = j | I \in U_t] = \frac{G(C_j^*; \mu_{1:t})}{G(C_{U_t}^*; \mu_{1:t})}$$

$$\mathbb{E}[G(C_I^*; \mu_{1:t}) | \mu_{1:t}, I \in U_t] = \sum_{j \in U_t} \frac{G(C_j^*; \mu_{1:t})}{G(C_{U_t}^*; \mu_{1:t})} G(C_j^*; \mu_{1:t}) \geq \frac{G(C_{U_t}^*; \mu_{1:t})^2 / |U_t|}{G(C_{U_t}^*; \mu_{1:t})} = \frac{G(C_{U_t}^*; \mu_{1:t})}{|U_t|}$$

Cauchy-Schwarz

Above the average

k -Means++

Lemma (Good iteration). Let $t \leq k - 1$. Conditioned on $\mu_{1:t}$ and on the event that μ_{t+1} lands in an uncovered cluster, $\mathbb{E}[\Phi_{t+1}] \leq \Phi_t$

Proof.

- Thus, we get that

$$\begin{aligned}\mathbb{E}[\Phi_{t+1} | \mu_{1:t}, I \in U_t] &\leq W_t \frac{G(C_{U_t}^*; \mu_{1:t}) - \mathbb{E}[G(C_I^*; \mu_{1:t}) | \mu_{1:t}, I \in U_t]}{|U_t| - 1} \\ &\leq W_t G(C_{U_t}^*; \mu_{1:t}) \frac{1 - 1/|U_t|}{|U_t| - 1} \\ &= \Phi_t\end{aligned}$$



k -Means++

Lemma (Bad iteration). Let $t \leq k - 1$. If μ_{t+1} lands in an already-hit cluster, then

$$\Phi_{t+1} - \Phi_t \leq \frac{G(C_{U_t}^*; \mu_{1:t})}{|U_t|}$$

Proof.

- In this case, $W_{t+1} = W_t + 1$, $U_{t+1} = U_t$
- Thus,

$$\Phi_{t+1} = W_{t+1} \frac{G(C_{U_{t+1}}^*; \mu_{1:t+1})}{|U_{t+1}|} = (W_t + 1) \frac{G(C_{U_t}^*; \mu_{1:t+1})}{|U_t|} \leq (W_t + 1) \frac{G(C_{U_t}^*; \mu_{1:t})}{|U_t|}$$



k -Means++

Lemma. For every $0 \leq t \leq k - 1$,

$$\mathbb{E}[\Phi_{t+1} - \Phi_t | \boldsymbol{\mu}_{1:t}] \leq \frac{G(C_{H_t}^*; \boldsymbol{\mu}_{1:t})}{k - t}$$

Proof.

- The probability that the $(t + 1)$ -th iteration is wasted is

$$p_{\text{waste}} = \Pr[\boldsymbol{\mu}_{t+1} \in C_{H_t}^* | \boldsymbol{\mu}_{1:t}] = \frac{G(C_{H_t}^*; \boldsymbol{\mu}_{1:t})}{G(\mathcal{X}; \boldsymbol{\mu}_{1:t})}$$

- Combining the **good iteration** and **bad iteration** lemmas, we have

$$\begin{aligned} \mathbb{E}[\Phi_{t+1} - \Phi_t | \boldsymbol{\mu}_{1:t}] &\leq 0 \cdot (1 - p_{\text{waste}}) + \frac{G(C_{U_t}^*; \boldsymbol{\mu}_{1:t})}{|U_t|} \cdot p_{\text{waste}} = \frac{G(C_{U_t}^*; \boldsymbol{\mu}_{1:t}) G(C_{H_t}^*; \boldsymbol{\mu}_{1:t})}{|U_t| G(\mathcal{X}; \boldsymbol{\mu}_{1:t})} \\ &\leq \frac{G(C_{H_t}^*; \boldsymbol{\mu}_{1:t})}{|U_t|} = \frac{G(C_{H_t}^*; \boldsymbol{\mu}_{1:t})}{k - |H_t|} \leq \frac{G(C_{H_t}^*; \boldsymbol{\mu}_{1:t})}{k - t} \end{aligned}$$



k -Means++

Theorem (Arthur-Vassilvitskii '07). Let $\mu_1, \dots, \mu_k \in \mathcal{X}$ be the output of k -means++ and $\mu_1^*, \dots, \mu_k^*$ be the optimal centers. Then,

$$\mathbb{E}[G(\mathcal{X}; \mu_{1:k})] \leq O(\log k) \cdot G(\mathcal{X}; \mu_{1:k}^*)$$

Proof.

$$\mathbb{E}[G(\mathcal{X}; \mu_{1:k})] = \mathbb{E}[G(C_{H_t}^*; \mu_{1:k})] + \sum_{t=0}^{k-1} \mathbb{E}[\Phi_{t+1} - \Phi_t] \leq 8G(\mathcal{X}; \mu_{1:k}^*) + \sum_{t=0}^{k-1} \mathbb{E}[\Phi_{t+1} - \Phi_t]$$

- By the total expectation, we have

$$\mathbb{E}[\Phi_{t+1} - \Phi_t] = \mathbb{E}[\mathbb{E}[\Phi_{t+1} - \Phi_t | \mu_{1:t}]] \leq \frac{\mathbb{E}[G(C_{H_t}^*; \mu_{1:t})]}{k-t} \leq \frac{8G(\mathcal{X}; \mu_{1:k}^*)}{k-t}$$

- Thus,

$$\sum_{t=0}^{k-1} \mathbb{E}[\Phi_{t+1} - \Phi_t] \leq 8G(\mathcal{X}; \mu_{1:k}^*) \cdot \sum_{t=1}^k \frac{1}{t} = O(\log k) \cdot G(\mathcal{X}; \mu_{1:k}^*)$$



k -Means++

k -means++ initialization:

1. Sample $\mu_1 \in \mathcal{X}$ uniformly at random
2. For each $j \in \{2, \dots, k\}$:
 - i. Sample $\mu_j = x \in \mathcal{X}$ with probability $\propto G(x; \mu_{1:j-1}) := \min_{i < j} \|x - \mu_i\|^2$

Theorem (Arthur-Vassilvitskii '07). Let $\mu_1, \dots, \mu_k \in \mathcal{X}$ be the output of k -means++ and $\mu_1^*, \dots, \mu_k^*$ be the optimal centers. Then,

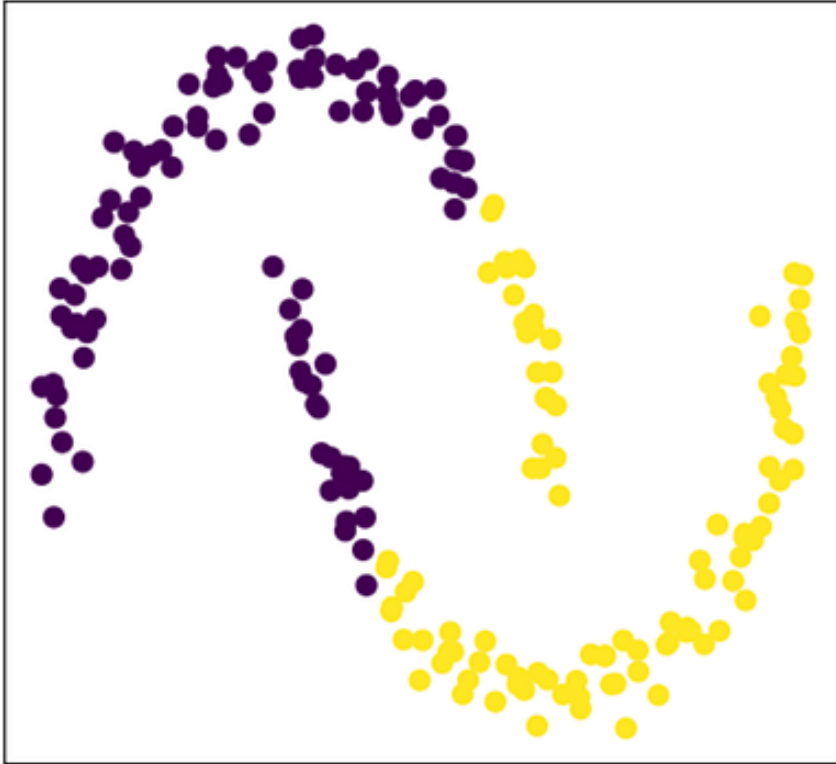
$$\mathbb{E}[G(\mathcal{X}; \mu_{1:k})] \leq O(\log k) \cdot G(\mathcal{X}; \mu_{1:k}^*)$$

- All the centers are from the input data
- Arthur-Vassilvitskii also proved that $O(\log k)$ -approximation of k -means++ is tight

Spectral Clustering

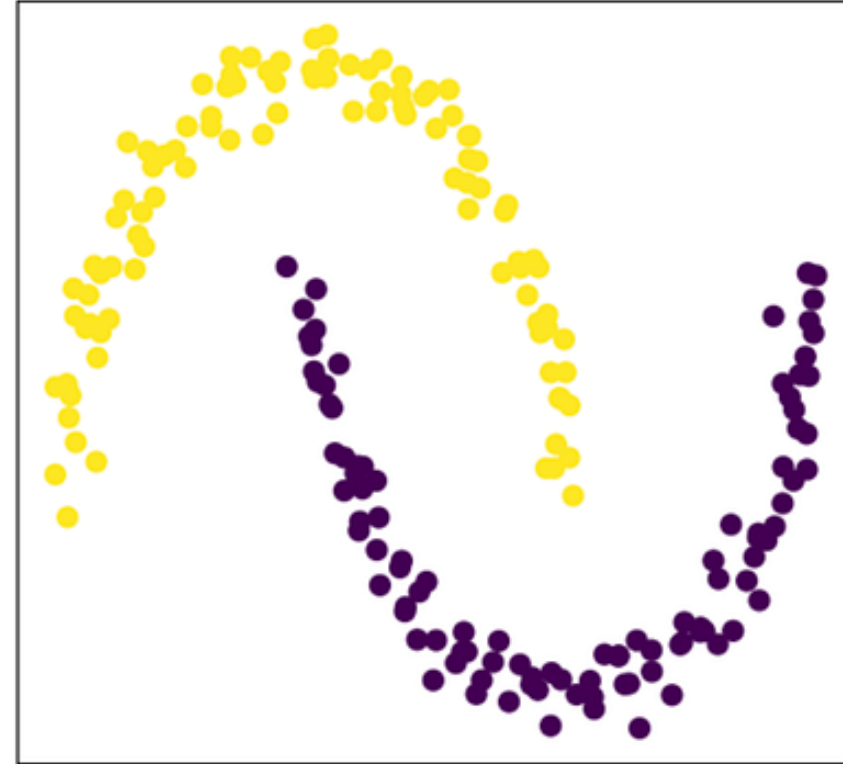
Comparing Clustering Algorithms on Non-Convex Data

K-Means Clustering (Fails)



Based on “closeness” to a center point

Spectral Clustering (Succeeds)

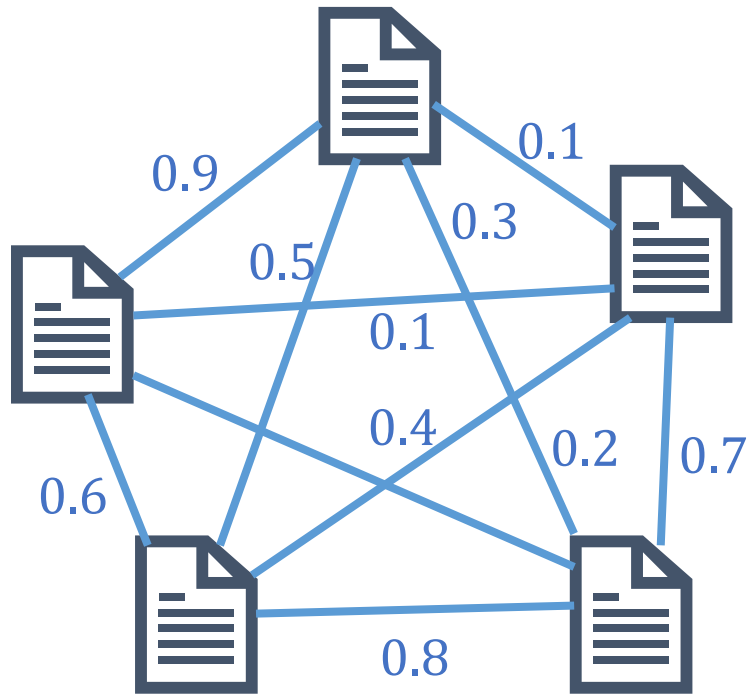


Based on “connectivity” or “similarity”

<https://dilipkumar.medium.com/spectral-clustering-algorithm-03a62854d19b>

Spectral Clustering

Convert data into a similarity graph



Similarity score $W(x_i, x_j)$

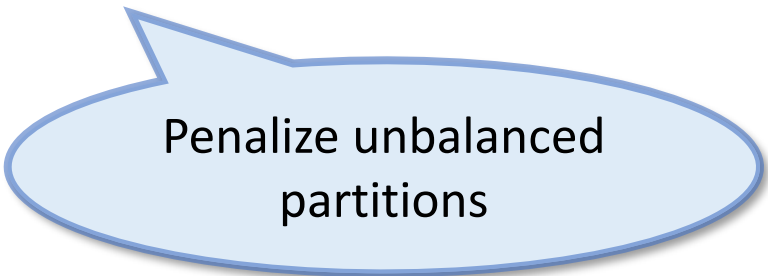


Spectral Clustering

Graph partitioning

- **Input:** a similarity graph $G = (V, E)$ with $w_{ij} = w_{ji} \geq 0$ and $w_{ii} = 0$
- **Output:** two nonempty sets A, B with $A \cup B = V$ and $A \cap B = \emptyset$
- **Goal:** Cut little total similarity while avoiding very small clusters

$$s(A, B) = \frac{\text{Cut}(A, B)}{|A||B|} = \frac{\sum_{i \in A, j \notin A} w_{ij}}{|A|(|V| - |A|)}$$

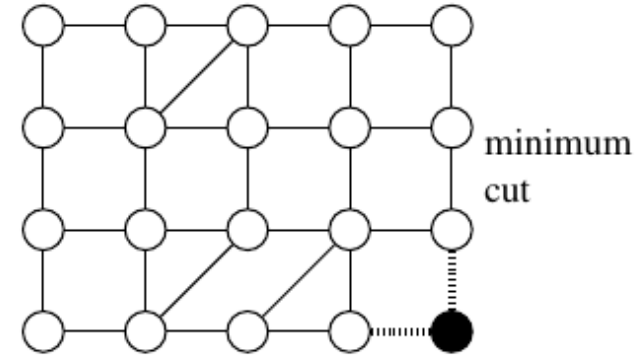
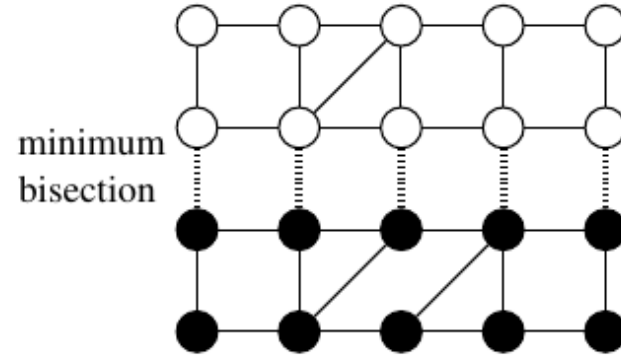


Penalize unbalanced partitions

Spectral Clustering

Cut objectives

$$\min_{|A|=|V|/2} \sum_{i \in A, j \notin A} w_{ij}$$

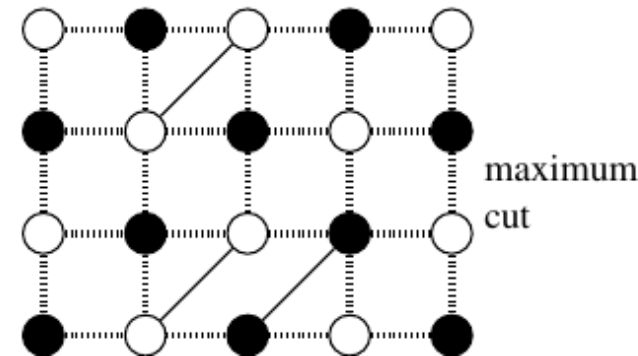
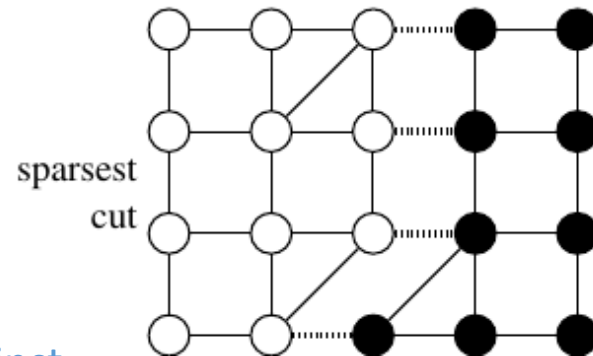


$$\min_A \sum_{i \in A, j \notin A} w_{ij}$$

may isolate a single vertex

$$\min_A \frac{\sum_{i \in A, j \notin A} w_{ij}}{|A|(|V| - |A|)}$$

trades crossing weight against balance



$$\max_A \sum_{i \in A, j \notin A} w_{ij}$$

Spectral Clustering

A partition as a vector

$$\mathbf{y} \in \mathbb{R}^n, \quad y_i = +1 \ (i \in A), \quad y_i = -1 \ (i \notin A)$$

For an edge $\{i, j\} \in E$:

- Same cluster: $y_i = y_j$, so $(y_i - y_j)^2 = 0$
- Different clusters: $y_i = -y_j$, so $(y_i - y_j)^2 = 4$

$$\text{Cut}(A, V - A) = w(E(A, V - A)) = \frac{1}{4} \sum_{\{i, j\} \in E} w_{ij} (y_i - y_j)^2$$

Spectral Clustering

Graph Laplacians

- Let $A = (w_{ij})_{i,j \in V}$ be the (weighted) **adjacency matrix** ($w_{ij} \in \{0,1\}$ for unweighted graph)
- Let $d_i = \sum_j w_{ij}$ be the (weighted) **degree** of the vertex $i \in V$, and $D = \text{diag}(d_1, \dots, d_n)$
- The **(combinatorial) Laplacian** is

$$L := D - A$$

- Then, for a cut indicator $y \in \{-1,1\}^n$,

$$\mathbf{y}^\top L \mathbf{y} = \sum_{i,j} L_{ij} y_i y_j = \sum_i d_i y_i^2 - \sum_{i,j} w_{ij} y_i y_j = \sum_i \sum_j w_{ij} (1 - y_i y_j) = \frac{1}{2} \sum_{i,j} w_{ij} (y_i - y_j)^2$$

- Thus, for $y \in \{-1,1\}^n$ indicating a cut $(A, V - A)$,

$$\mathbf{y}^\top L \mathbf{y} = 4 \cdot \text{Cut}(A, V - A)$$

Spectral Clustering

Properties of the Laplacian

- The Laplacian is symmetric and positive semidefinite:

$$L^T = (D - A)^T = D - A = L, \quad \mathbf{y}^T L \mathbf{y} = \sum_{\{i,j\} \in E} w_{ij} (y_i - y_j)^2 \geq 0$$

- Every row sums to zero, so $L\mathbf{1} = \mathbf{0}$:

$$\sum_j (D - A)_{ij} = d_i - \sum_j w_{ij} = d_i - d_i = 0$$

- The smallest eigenvalue $\lambda_1 = 0$, and the corresponding eigenvector $\mathbf{v}_1 = \frac{1}{\sqrt{n}} \mathbf{1}$
- If the positive-weight edges connect the graph, then $\lambda_2 > 0$. Moreover, every connected component contributes one zero eigenvalue

Spectral Clustering

If the positive-weight edges connect the graph, then $\lambda_2 > 0$. Moreover, every connected component contributes one zero eigenvalue

$$\mathbf{y}^\top L \mathbf{y} = \sum_{\{i,j\} \in E} w_{ij} (y_i - y_j)^2 \geq 0$$

- For every connected component $C \subset V$, let $\mathbf{v}_C := \frac{1}{\sqrt{|C|}} \mathbf{1}_C$
- We have $\mathbf{v}_C^\top L \mathbf{v}_C = 0$, and $\mathbf{v}_C \perp \mathbf{v}_{C'}$ for different components $C, C' \subset V$
- Thus, $\text{span}\{\mathbf{v}_C\} \subseteq \ker(L)$

Spectral Clustering

If the positive-weight edges connect the graph, then $\lambda_2 > 0$. Moreover, every connected component contributes one zero eigenvalue

$$\mathbf{y}^\top L \mathbf{y} = \sum_{\{i,j\} \in E} w_{ij} (y_i - y_j)^2 \geq 0$$

- Conversely, suppose $\mathbf{y}^\top L \mathbf{y} = 0$ for some $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{y} \neq \mathbf{0}$
- $A_+ = \{i \in V : y_i > 0\}$, $A_0 = \{i \in V : y_i = 0\}$, and $A_- = \{i \in V : y_i < 0\}$
- Each of A_+ , A_- , A_0 is a union of connected components
- The entries of $\mathbf{y}$ in the same component have the same value
- Thus, $\mathbf{y} \in \text{span}\{\mathbf{v}_C\}$

Spectral Clustering

Minimum bisection

$$\min_{|A|=|V|/2} \text{cut}(A, V - A)$$

- Equivalently,

$$\min \mathbf{y}^\top L \mathbf{y}, \quad \mathbf{y} \in \{-1, 1\}^n, \quad \mathbf{y}^\top \mathbf{1} = 0$$

- This is an **integer programming** problem, which is **NP-hard**
- Consider the following **relaxation**:

$$\min \mathbf{y}^\top L \mathbf{y}, \quad \mathbf{y} \in \mathbb{R}^n, \quad \mathbf{y}^\top \mathbf{y} = n, \quad \mathbf{y}^\top \mathbf{1} = 0$$

- The feasible region of the relaxed problem contains the optimal integral solution

Spectral Clustering

The Fiedler vector

For a connected graph, let $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n$ be the eigenvalues of L with orthonormal eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$. Then

$$\lambda_2 = \min_{\mathbf{y} \in \mathbb{R}^n: \mathbf{y} \perp \mathbf{1}} \frac{\mathbf{y}^\top L \mathbf{y}}{\mathbf{y}^\top \mathbf{y}}$$

Proof.

- Write $\mathbf{y} = \sum_{i=2}^n \alpha_i \mathbf{v}_i$

$$\frac{\mathbf{y}^\top L \mathbf{y}}{\mathbf{y}^\top \mathbf{y}} = \frac{\sum_{i=2}^n \lambda_i \alpha_i^2}{\sum_{i=2}^n \alpha_i^2} \geq \frac{\left(\min_{2 \leq i \leq n} \lambda_i \right) \cdot \sum_{i=2}^n \alpha_i^2}{\sum_{i=2}^n \alpha_i^2} = \lambda_2$$

- Equality holds when $\mathbf{y} = \mathbf{v}_2$
- We call $\mathbf{v}_2$ the **Fiedler vector**

Spectral Clustering

Spectral partitioning algorithm (ver. 1)

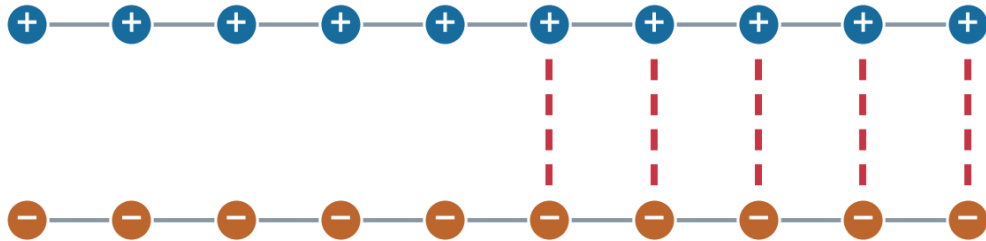
1. Compute v_2 , an eigenvector for the second-smallest eigenvalue of L
2. Round v_2 by median partition:
 - i. Sort its coordinates: $(v_2)_{\pi(1)} \leq \dots \leq (v_2)_{\pi(n)}$
 - ii. Let $A = \{\pi(1), \dots, \pi(n/2)\} \subset V$
 - iii. Return the cut $(A, V - A)$

Spectral Clustering

Guattery–Miller '98:

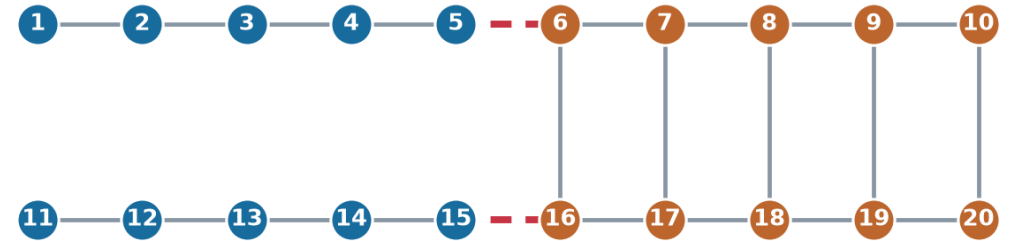
A Median partition: 5 cut edges

All upper entries are positive; all lower entries are negative.



B Optimal bisection: 2 cut edges

Both antennae form one part; the ladder body forms the other.



- The approximation ratio for minimum bisection can be $\Omega(n)$

Spectral Clustering

Spectral partitioning algorithm (ver. 2)

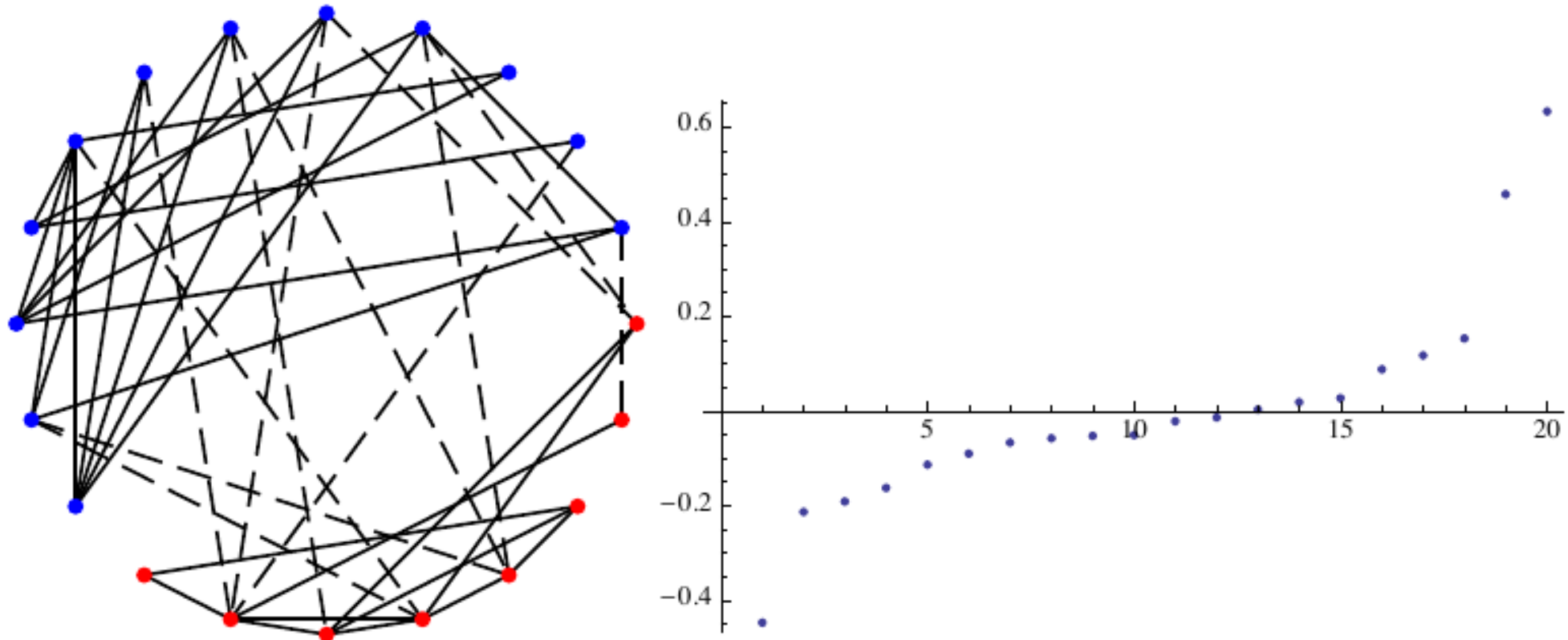
1. Compute v_2 , an eigenvector for the second-smallest eigenvalue of L
2. Round v_2 with a sweep cut:
 - i. Sort its coordinates: $(v_2)_{\pi(1)} \leq \dots \leq (v_2)_{\pi(n)}$
 - ii. For each $i = 1, \dots, n - 1$, let $A_i = \{\pi(1), \dots, \pi(i)\} \subset V$
 - iii. Return the cut minimizing $\text{cut}(A_i, V - A_i) / (|A_i| |V - A_i|)$

- The threshold is chosen by a sweep over candidate cuts
- The resulting partition need not be an exact bisection
- We will see that this algorithm produces an approximation of the sparsest cut problem

Spectral Clustering

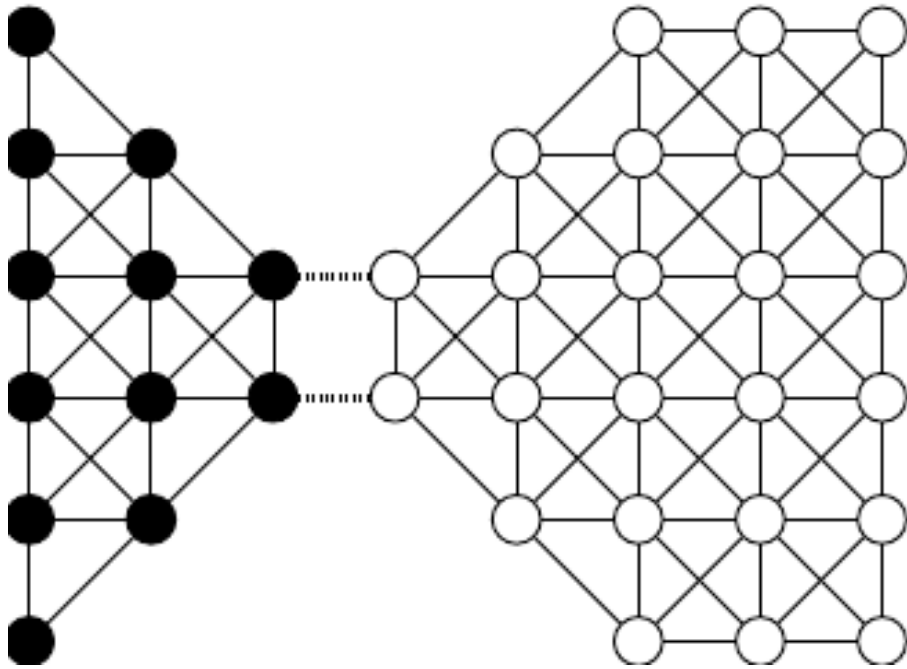
Sweep cuts

- Each coordinate of $\mathbf{v}_2$ assigns a real value to one vertex

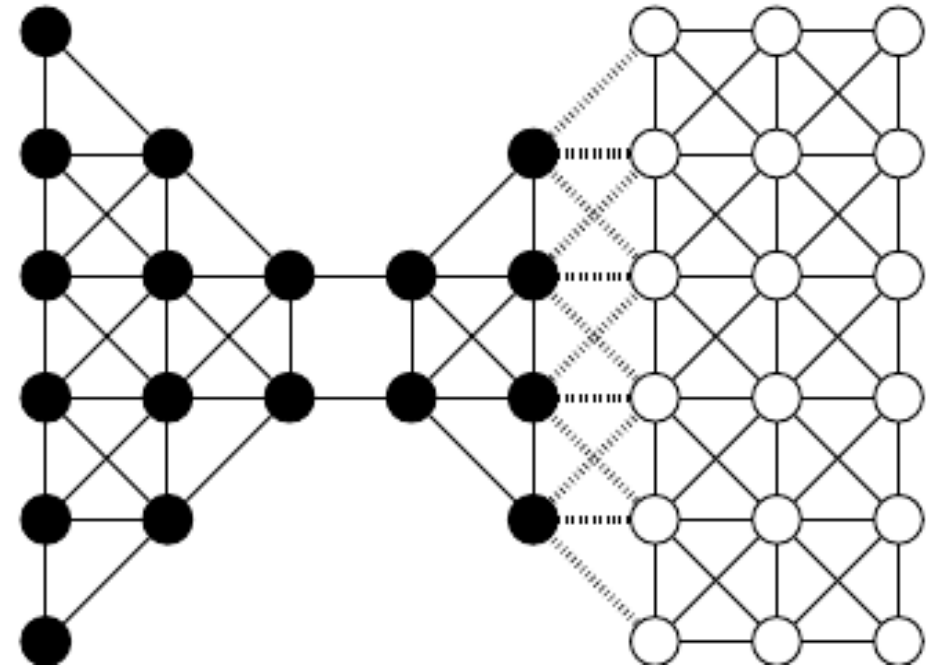


Spectral Clustering

Balance after rounding



Slightly unbalanced cut



Balanced cut

Spectral Clustering

Ratio Cut

$$\min_{A \subset V} |V| \cdot \frac{\text{cut}(A, V - A)}{|A||V - A|} = \left(\frac{\text{cut}(A, V - A)}{|A|} + \frac{\text{cut}(A, V - A)}{|V - A|} \right)$$

- Recall that $\text{cut}(A, V - A) = \frac{1}{4} \mathbf{y}^\top L \mathbf{y}$ for $y_i = 1$ if $i \in A$ and $y_i = -1$ if $i \notin A$
- If we set $y_i = \frac{1}{|A|}$ if $i \in A$ and $y_i = -\frac{1}{|V - A|}$ if $i \notin A$, then

$$\mathbf{y}^\top \mathbf{1} = |A| \cdot \frac{1}{|A|} - |V - A| \cdot \frac{1}{|V - A|} = 0$$

$$\mathbf{y}^\top L \mathbf{y} = \sum_{\{i,j\} \in E} w_{ij} (y_i - y_j)^2 = \sum_{i \in A, j \notin A} w_{ij} \left(\frac{1}{|A|} + \frac{1}{|V - A|} \right)^2 = \text{cut}(A, V - A) \left(\frac{1}{|A|} + \frac{1}{|V - A|} \right)^2$$

$$\mathbf{y}^\top \mathbf{y} = \frac{1}{|A|} + \frac{1}{|V - A|}$$

Spectral Clustering

Ratio Cut

$$\min_{A \subset V} |V| \cdot \frac{\text{cut}(A, V - A)}{|A||V - A|} = \left(\frac{\text{cut}(A, V - A)}{|A|} + \frac{\text{cut}(A, V - A)}{|V - A|} \right)$$

- Recall that $\text{cut}(A, V - A) = \frac{1}{4} \mathbf{y}^\top L \mathbf{y}$ for $y_i = 1$ if $i \in A$ and $y_i = -1$ if $i \notin A$
- If we set $y_i = \frac{1}{|A|}$ if $i \in A$ and $y_i = -\frac{1}{|V-A|}$ if $i \notin A$, then

$$\frac{\mathbf{y}^\top L \mathbf{y}}{\mathbf{y}^\top \mathbf{y}} = \text{RCut}(A), \quad \mathbf{y} \perp \mathbf{1}$$

- Therefore, the Fiedler vector is a relaxation of the ratio cut:

$$\lambda_2 = \min_{\mathbf{y} \in \mathbb{R}^n: \mathbf{y} \perp \mathbf{1}} \frac{\mathbf{y}^\top L \mathbf{y}}{\mathbf{y}^\top \mathbf{y}} \leq \text{OPT}_{\text{RCut}}(G)$$

Spectral Clustering

Isoperimetric ratio

- Let $G = (V, E)$ be an undirected d -regular graph
- The isoperimetric ratio of a cut $(A, V - A)$ is defined as

$$h(A) := \frac{\text{Cut}(A, V - A)}{\min\{|A|, |V - A|\}}$$

- It is a 2-approximation of the ratio cut:

$$h(A) \leq \text{RCut}(A) \leq 2h(A)$$

Theorem (Cheeger, ver. 1). Let $h_G := \min_{A \subset V} h(A)$ be the isoperimetric parameter of G . Then

$$\frac{\lambda_2}{2} \leq h_G \leq \sqrt{2d_{\max}\lambda_2}, \quad d_{\max} := \max_{i \in V} d_i$$

Spectral Clustering

- Let $\mathbf{v}_2$ be the Fiedler vector, i.e., $L\mathbf{v}_2 = \lambda_2\mathbf{v}_2$. We may choose $\mathbf{v}_2$ so that the number of positive entries are at most $n/2$
- Let $\mathbf{x} := \max\{\mathbf{v}_2, \mathbf{0}\}$ (entry-wise)
- We have $\mathbf{x}^\top \mathbf{x} = \mathbf{x}^\top \mathbf{v}_2$ and $\mathbf{x}^\top L(\mathbf{x} - \mathbf{v}_2) \leq 0$
- Thus,

(+) · (−) · (+)

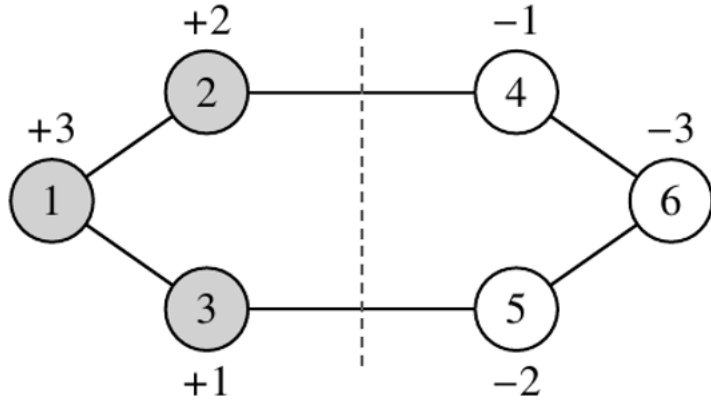
$$\mathbf{x}^\top L\mathbf{x} = \mathbf{x}^\top L(\mathbf{x} - \mathbf{v}_2) + \mathbf{x}^\top L\mathbf{v}_2 \leq \mathbf{x}^\top L\mathbf{v}_2 = \lambda_2 \cdot \mathbf{x}^\top \mathbf{v}_2 = \lambda_2 \cdot \mathbf{x}^\top \mathbf{x}$$

We will run the sweep cut for $\mathbf{x}$ and will show that there is a good cut in the **positive part** of the Fiedler vector

- Wlog, we may assume that the positive entries of $\mathbf{x}$ occur in a descending order
- Define the **positive-threshold sweep cut expansion**:

$$h := \min_{i \in [n]: x_i > 0} \frac{\text{Cut}([i], [n] - [i])}{\min\{i, n - i\}} = \min_{i \in [n]: x_i > 0} \frac{\text{Cut}([i], [n] - [i])}{i} \geq h_G$$

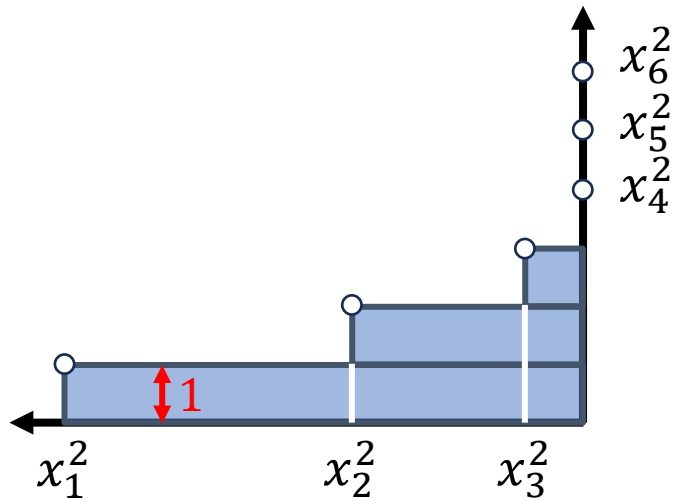
Spectral Clustering



$$\mathbf{v}_2 = (3, 2, 1, -1, -2, -3)$$

$$\lambda_2 = 1$$

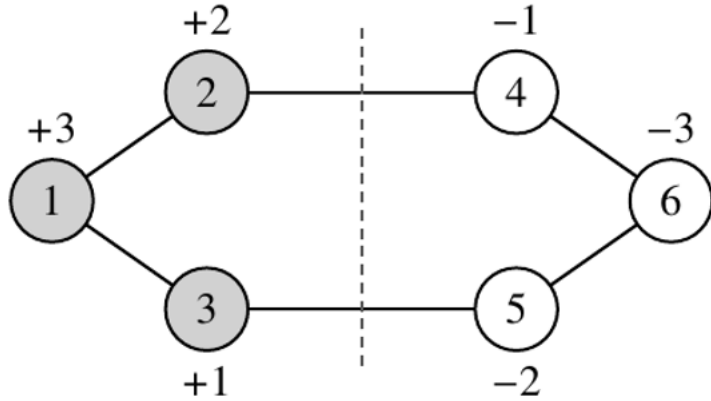
$$\mathbf{x} = (3, 2, 1, 0, 0, 0)$$



By double-counting, we have

$$\sum_{i:x_i>0} i(x_i^2 - x_{i+1}^2) = \sum_{i:x_i>0} x_i^2 = \mathbf{x}^\top \mathbf{x}$$

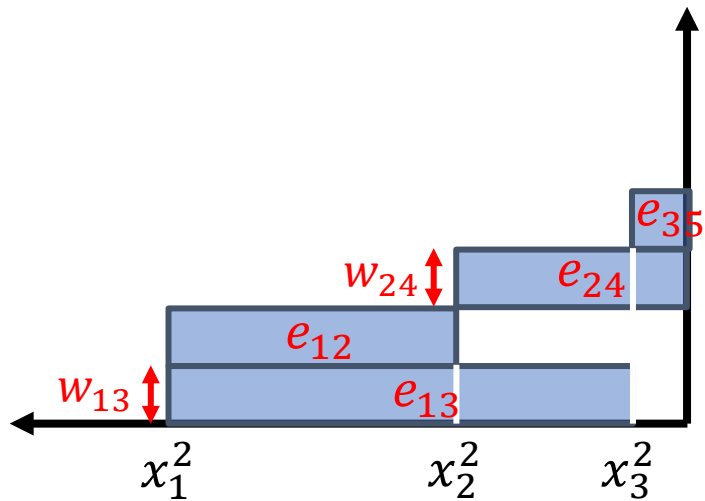
Spectral Clustering



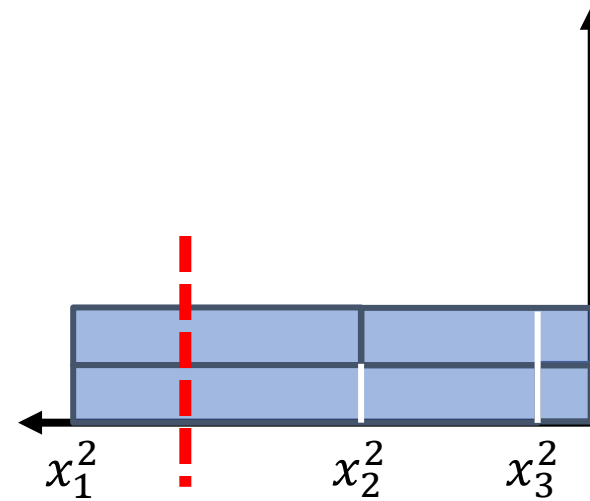
$$\mathbf{v}_2 = (3, 2, 1, -1, -2, -3)$$

$$\lambda_2 = 1$$

$$\mathbf{x} = (3, 2, 1, 0, 0, 0)$$



Vertically
compactify

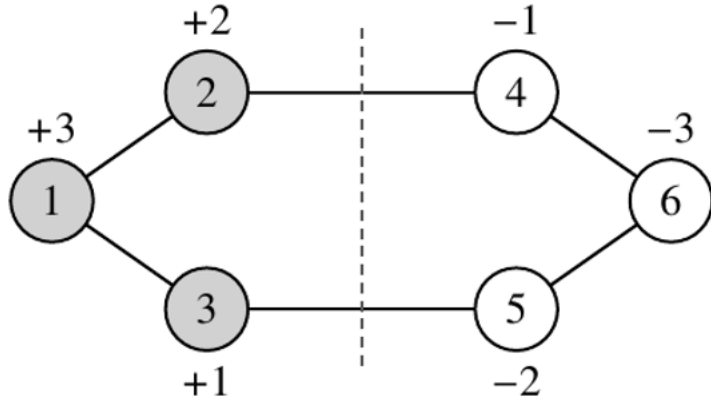


$$\text{Cut}(\{1\})(x_1^2 - x_2^2)$$

$$\text{Cut}(\{1,2\})(x_2^2 - x_3^2)$$

$$\text{Cut}(\{1,2,3\})(x_3^2 - x_4^2)$$

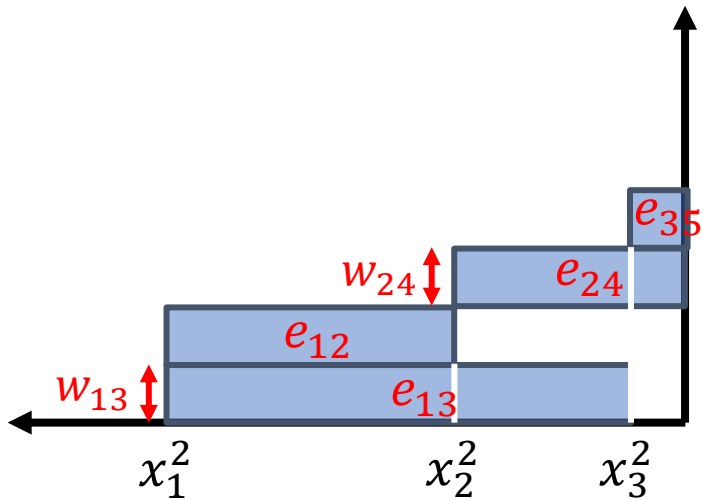
Spectral Clustering



$$\mathbf{v}_2 = (3, 2, 1, -1, -2, -3)$$

$$\lambda_2 = 1$$

$$\mathbf{x} = (3, 2, 1, 0, 0, 0)$$



By double-counting, we have

$$\sum_{i: x_i > 0} \text{Cut}([i], [n] - [i]) (x_i^2 - x_{i+1}^2) = \sum_{\{i,j\} \in E} w_{ij} |x_i^2 - x_j^2|$$

Spectral Clustering

$$h := \min_{i \in [n]: x_i > 0} \frac{\text{Cut}([i], [n] - [i])}{\min\{i, n - i\}} = \min_{i \in [n]: x_i > 0} \frac{\text{Cut}([i], [n] - [i])}{i}$$

$$h \cdot i \leq \text{Cut}([i], [n] - [i]), \quad \forall i \in [n] : x_i > 0$$

$$\begin{aligned} \sum_{\{i,j\} \in E} w_{ij} |x_i^2 - x_j^2| &= \sum_{i: x_i > 0} \text{Cut}([i], [n] - [i]) (x_i^2 - x_{i+1}^2) \\ &\geq \sum_{i \in [n]: x_i > 0} h \cdot i (x_i^2 - x_{i+1}^2) \\ &= h \mathbf{x}^\top \mathbf{x} \end{aligned}$$

Spectral Clustering

$$\begin{aligned}\sum_{\{i,j\} \in E} w_{ij} |x_i^2 - x_j^2| &= \sum_{\{i,j\} \in E} w_{ij} |x_i - x_j| (x_i + x_j) \\ &\leq \left(\sum_{\{i,j\} \in E} w_{ij} (x_i - x_j)^2 \right)^{1/2} \left(\sum_{\{i,j\} \in E} w_{ij} (x_i + x_j)^2 \right)^{1/2} \\ &= \sqrt{\mathbf{x}^\top L \mathbf{x}} \left(\sum_{\{i,j\} \in E} w_{ij} (x_i + x_j)^2 \right)^{1/2} \\ &\leq \sqrt{\mathbf{x}^\top L \mathbf{x}} \left(\sum_{\{i,j\} \in E} 2w_{ij} (x_i^2 + x_j^2) \right)^{1/2} \\ &= \sqrt{\mathbf{x}^\top L \mathbf{x}} \left(2 \sum_i x_i^2 \sum_{j: \{i,j\} \in E} w_{ij} \right)^{1/2} \\ &= \sqrt{\mathbf{x}^\top L \mathbf{x}} \sqrt{2 \mathbf{x}^\top D \mathbf{x}}\end{aligned}$$

Spectral Clustering

$$h\mathbf{x}^\top \mathbf{x} \leq \sum_{\{i,j\} \in E} w_{ij} |x_i^2 - x_j^2| \leq \sqrt{\mathbf{x}^\top L \mathbf{x}} \sqrt{2\mathbf{x}^\top D \mathbf{x}} = \sqrt{\lambda_2 \mathbf{x}^\top \mathbf{x}} \sqrt{2\mathbf{x}^\top D \mathbf{x}}$$

$$h \leq \sqrt{2\lambda_2} \sqrt{\frac{\mathbf{x}^\top D \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}} \leq \sqrt{2\lambda_2 d_{\max}}$$

$$h_G \leq \min_{i \in [n]} \frac{\text{Cut}([i], [n] - [i])}{\min\{i, n - i\}} \leq h \leq \sqrt{2\lambda_2 d_{\max}}$$



Spectral Clustering

Spectral partitioning algorithm (ver. 2)

1. Compute $\mathbf{v}_2$, an eigenvector for the second-smallest eigenvalue of L
2. Round $\mathbf{v}_2$ with a sweep cut:
 - i. Sort its coordinates: $(\mathbf{v}_2)_{\pi(1)} \leq \dots \leq (\mathbf{v}_2)_{\pi(n)}$
 - ii. For each $i = 1, \dots, n - 1$, let $A_i = \{\pi(1), \dots, \pi(i)\} \subset V$
 - iii. Return the cut minimizing $\text{cut}(A_i, V - A_i) / (|A_i| |V - A_i|)$

$$\lambda_2 \leq \text{OPT}_{\text{RCut}}(G) \leq \text{RCut}(A_i) \leq 2h(A_i) \leq 2\sqrt{2\lambda_2 d_{\max}}$$

Spectral Clustering

(Uniform) Sparsest Cut

$$\min_{A \subset V} \frac{\text{cut}(A, V - A)}{\text{vol}(A)\text{vol}(V - A)} \propto \left(\frac{\text{cut}(A, V - A)}{\text{vol}(A)} + \frac{\text{cut}(A, V - A)}{\text{vol}(V - A)} \right)$$

$$\text{vol}(A) := \sum_{i \in A} d_i$$

- Similar to the ratio cut, if we define $y_i = \frac{1}{\text{vol}(A)}$ for $i \in A$ and $y_i = -\frac{1}{\text{vol}(V-A)}$ for $i \notin A$, then

$$\mathbf{1}^\top D \mathbf{y} = \mathbf{0}$$

$$\frac{\mathbf{y}^\top L \mathbf{y}}{\mathbf{y}^\top D \mathbf{y}} = \frac{\sum_{\{i,j\} \in E} w_{ij} (1/\text{vol}(A) - 1/\text{vol}(V - A))^2}{\sum_{i \in A} d_i / \text{vol}(A)^2 + \sum_{i \notin A} d_i / \text{vol}(V - A)^2} = \text{cut}(A, V - A) \left(\frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(V - A)} \right)$$

Spectral Clustering

(Uniform) Sparsest Cut

$$\min_{A \subset V} \frac{\text{cut}(A, V - A)}{\text{vol}(A)\text{vol}(V - A)} \propto \left(\frac{\text{cut}(A, V - A)}{\text{vol}(A)} + \frac{\text{cut}(A, V - A)}{\text{vol}(V - A)} \right)$$

$$\text{vol}(A) := \sum_{i \in A} d_i$$

- Thus, the relaxation is the following generalized eigenvalue problem:

$$\min_{\mathbf{y} \neq 0, \mathbf{1}^\top D \mathbf{y} = 0} \frac{\mathbf{y}^\top L \mathbf{y}}{\mathbf{y}^\top D \mathbf{y}}, \quad L \mathbf{v}_2 = \lambda_2 D \mathbf{v}_2$$

- Let $\mathbf{z} := D^{1/2} \mathbf{y}$

$$\min_{\mathbf{z} \perp \mathbf{d}^{1/2}} \frac{\mathbf{z}^\top (D^{-1/2} L D^{-1/2}) \mathbf{z}}{\mathbf{z}^\top \mathbf{z}}, \quad \mathbf{d} = D \mathbf{1}$$

Spectral Clustering

Normalized Laplacian

$$N = D^{-1/2}LD^{-1/2} = I - D^{-1/2}AD^{-1/2}$$

- The eigenvalues of N are $0 = v_1 \leq v_2 \leq \dots \leq v_n$

$$v_2 = \min_{\mathbf{y} \perp \mathbf{d}} \frac{\mathbf{y}^\top L \mathbf{y}}{\mathbf{y}^\top D \mathbf{y}}$$

Conductance

$$\phi(A) = \frac{\text{cut}(A, V - A)}{\min\{\text{vol}(A), \text{vol}(V - A)\}}, \quad \phi_G = \min_{A \subset V} \phi(A)$$

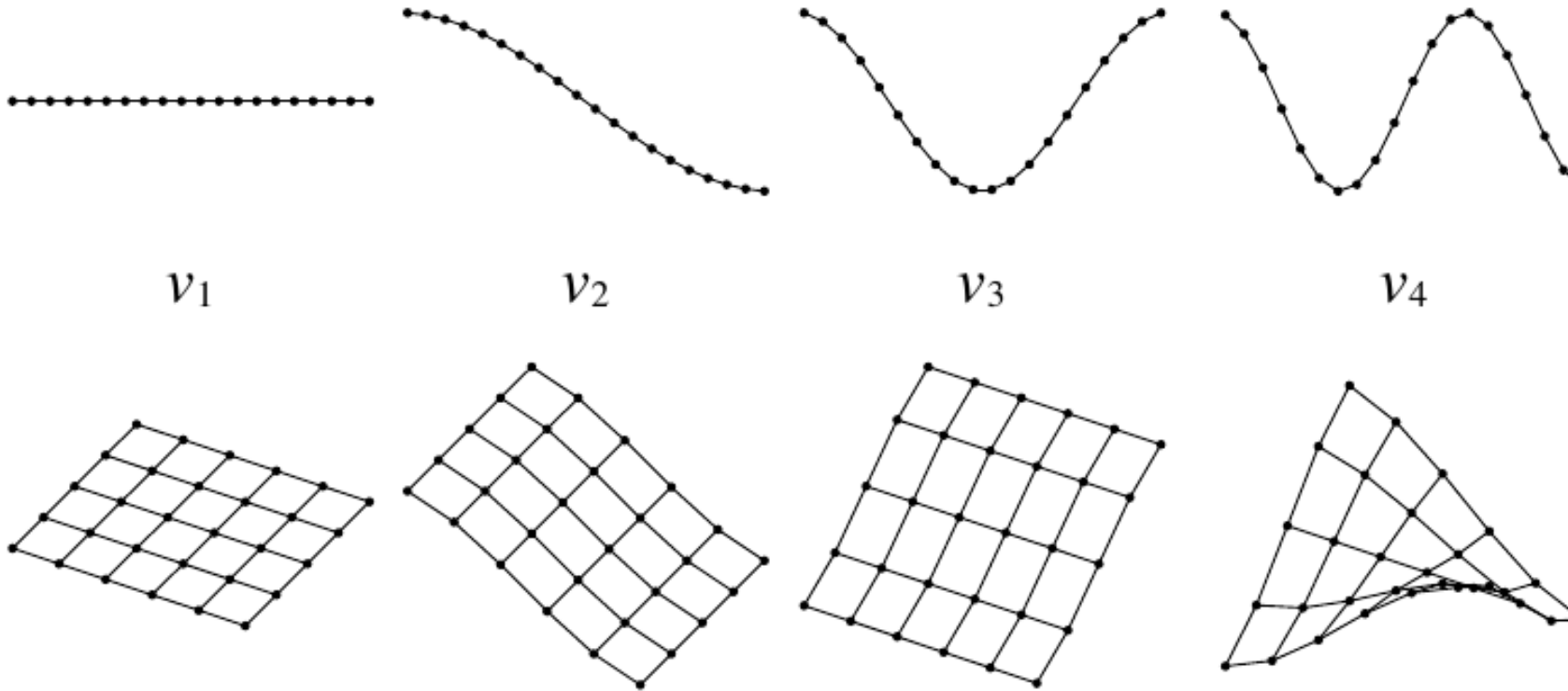
Theorem (Cheeger, ver. 2). For any graph G ,

$$v_2/2 \leq \phi_G \leq \sqrt{2v_2}$$

Spectral Clustering

Intuition from physics:

- Edges are springs, and vertices are point masses



Vibrational modes:

$$x_i(t) = v_i \cos(\omega t)$$

Hooke's law:

$$F_i = - \sum_j w_{ij} (x_i - x_j) \\ = -(Lx)_i$$

Newton's law:

$$M\ddot{x} = -Lx$$

$$Lv = \omega^2 Mv$$

Spectral Clustering

Image segmentation

- Each pixel i has a location $\mathbf{p}_i$ and brightness b_i , regarded as a vertex in the image graph
- Gaussian kernel for similarity:

$$w_{ij} = \exp\left(-\frac{\|\mathbf{p}_i - \mathbf{p}_j\|^2}{\alpha} - \frac{|b_i - b_j|^2}{\beta}\right), \quad \forall i \neq j$$

- Set $w_{ii} = 0$ and omit edges between pixels that are far apart
- Recursively partition the image graph

Spectral Clustering

Sparsest cut for image segmentation

Input photograph



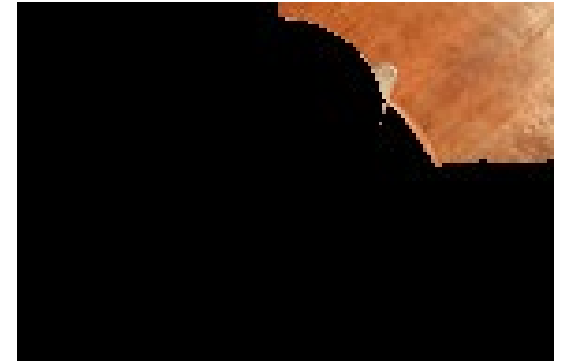
Region 1 (25.0%)



Region 2 (24.5%)



Region 3 (15.4%)



Region 4 (14.6%)



Region 5 (9.2%)



Region 6 (7.8%)



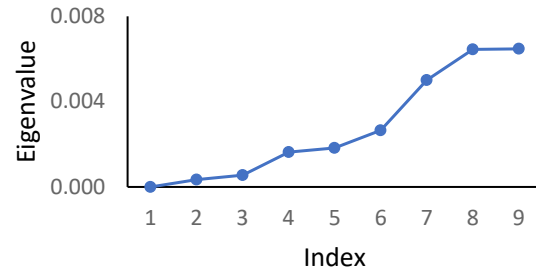
Region 7 (3.6%)



144 × 96 pixels, radius 5, $\alpha = 16$, $\beta = 0.01$, minimum region size 484 pixels

Spectral Clustering

Eigenvectors of the image graph



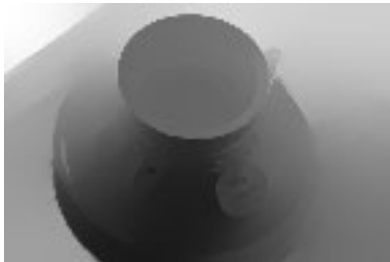
y_2 ($\lambda_2 = 0.00035$)



y_3 ($\lambda_3 = 0.00055$)



y_4 ($\lambda_4 = 0.00163$)



y_5 ($\lambda_5 = 0.00183$)



y_6 ($\lambda_6 = 0.00266$)



y_7 ($\lambda_7 = 0.00501$)



y_8 ($\lambda_8 = 0.00645$)



y_9 ($\lambda_9 = 0.00647$)

